

RWR GROUP

SLACK LMS PLATFORM

Software Engineering & Design Document

v1.0 · April 2026 · Rapid Application Development

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Companion Docs	WAA Tech Doc v1.5 · CMG Tech Doc v2.1 · Master System Prompt v2.0 · Skills Compendium March 2026

PURPOSE

This document is the complete software engineering and design reference for the RWR Slack LMS platform. It covers the full lifecycle from initiation through deployment and testing, following Rapid Application Development (RAD) methodology. It includes: use cases, user stories, UML activity diagrams, database schema, data dictionary, CSV specifications, workflow designs, skills compendium, deployment architecture, and project management plan.

1. Project Overview & RAD Methodology

1.1 Project Charter

Property	Value
Project Name	RWR Slack LMS Platform — Recruitment Training System
Project Sponsor	RWR Group (RWR Health, Hospoworld, Retailworld, RWR Construction)
Target Markets	New Zealand and Australia
Platform Tagline	"We don't recruit — we empower those who do."
Objective	Deliver a 288-lesson, 12-month recruitment training curriculum via Slack micro-learning, with a 14-agent AI content factory and a GAS+Sheets+AppSheet automation platform
Learner Target	Recruiting professionals across 4 specialist brands; 50 initial users, scalable to 500
Content Scale	288 lessons across 12 months (M01–M12) · 4 weeks/month · 5 daily micro + 1 weekly deep per week
Methodology	Rapid Application Development (RAD) — 4 phases with iterative build-test cycles
Start Date	March 2026
Production Target	April 2026 (Phase 1 live); full 288-lesson library Q3 2026

1.2 RAD Phase Model

The platform follows a 4-phase RAD model. Each phase produces working software, not just documentation.

Phase	Name	Duration	Deliverables	Gate
Phase 1	Requirements Planning	Weeks 1–2	Project charter · stakeholder analysis · use cases · user stories · data model design · skills compendium · CMC approval	Operator sign-off on Final Output Package (CMC Step 7)
Phase 2	User Design — Content Engine	Weeks 3–6	14-agent content factory · Master System Prompt v2.0 · ULC template · SOP-05 rubric · 288-lesson content plan · Notion curriculum database · Publish Panel MVP	QA pipeline produces first STRONG_PASS lesson; Publish Panel gates working
Phase 3	User Design — Delivery Engine	Weeks 7–10	GAS codebase (00_–21_) · SheetDb schema · 3 slash commands live · AppSheet admin console · Slack Workflow Builder admin workflows · test harness · smoke tests pass	All 8 smoke tests PASS; UAT sign-off from HR Manager + LMS Admin
Phase 4	Cutover & Deployment	Weeks 11–12	Production deployment · Script Properties set · triggers active · Lessons.csv loaded · pilot cohort enrolled · monitoring live · RUNBOOK.html delivered	All SMOKE-01–08 pass in production; pilot cohort receives first lesson

1.3 Stakeholder Register

Stakeholder	Role	Involvement	Authority
LMS Platform Lead	Final decision authority on all production automations	Checkpoint 2 sign-off; deployment approval	Highest — overrides all
Platform Architect	Technical design validation; Firebase + AI Studio config	Checkpoint 2 sign-off; skill mapping review	Technical authority

Stakeholder	Role	Involvement	Authority
Content Operations	CMC workflow owner; signs off lesson generation plans	Step 7 Final Output Package approval	CMC authority
System Administrator	Script Properties; credential rotation; deployment ops	All deployment steps; secret rotation	Operations authority
HR Manager	Primary admin user; enrolls learners; monitors progress	UAT-01, UAT-03, UAT-04, UAT-06; go/no-go on learner experience	Business authority
LMS Administrator	Content approval; QA dashboard; system status monitoring	UAT-06; AppSheet admin console testing	Admin authority
Learner (Recruiting Professional)	End user; receives and submits lessons via Slack	UAT-02, UAT-05; feedback on lesson quality and delivery	Consumer
QA Reviewer (AI Agent)	Final content gate; issues binding SOP-05 verdicts	All lesson approvals; FAIL routing	Highest content authority
Pipeline Orchestrator (AI)	Conducts 14-agent pipeline; cannot skip Hard FAIL gates	All content pipeline runs	Highest pipeline authority
Manager (Direct Report Leader)	Receives gate-outcome notifications; enrolls direct reports	Manager role Phase 2; /team command	Line management authority

1.4 RAD Project Timeline — Milestones

Milestone	Phase	Target Date	Owner	Acceptance Criteria
M1: Master System Prompt v2.0 finalised	P1	Week 1	Content Operations	All 14 agents specified; authority hierarchy documented; pipeline sequence complete
M2: 41-column Lessons.csv canonical schema locked	P1	Week 1	Platform Architect	Schema contract signed; all field names, types, owners, constraints documented
M3: CMC Final Output Package approved	P1	Week 2	Operator	All 5 CMC outputs present; Content Agent not triggered before this
M4: First lesson passes full 14-agent pipeline	P2	Week 4	Content Operations	QA verdict PASS or STRONG_PASS; Submit Block auto-generated; csv-qa-sync writes to Sheets
M5: 24 lessons (Month 1) in Notion — Ready status	P2	Week 6	Content Operations	All M01 lessons QA-approved; SlackThreadText populated; Status=Ready
M6: GAS codebase live — 3 slash commands responding	P3	Week 7	Platform Architect	SMOKE-01, SMOKE-02, SMOKE-03 pass in staging
M7: AppSheet admin console operational	P3	Week 8	System Administrator	HR Manager completes UAT-04 without developer help
M8: All 8 smoke tests PASS in staging	P3	Week 10	System Administrator	Full smoke test suite green; no CRITICAL or WARNING latency breaches
M9: UAT sign-off obtained	P3	Week 10	LMS Platform Lead	All 4 UAT participant roles signed; 0 blocking defects
M10: Production deployment complete	P4	Week 11	Platform Architect	Deployment checklist all ☒ ; Script Properties set; triggers active

Milestone	Phase	Target Date	Owner	Acceptance Criteria
M11: Pilot cohort (5 learners) enrolled and active	P4	Week 12	HR Manager	All 5 learners receive Day 1 lesson; at least 3 submit; no delivery errors
M12: Full 288-lesson library complete	P4	Q3 2026	Content Operations	All 288 lessons Status=Ready or Live; all SOP-05 scores ≥ 60

2. System Architecture

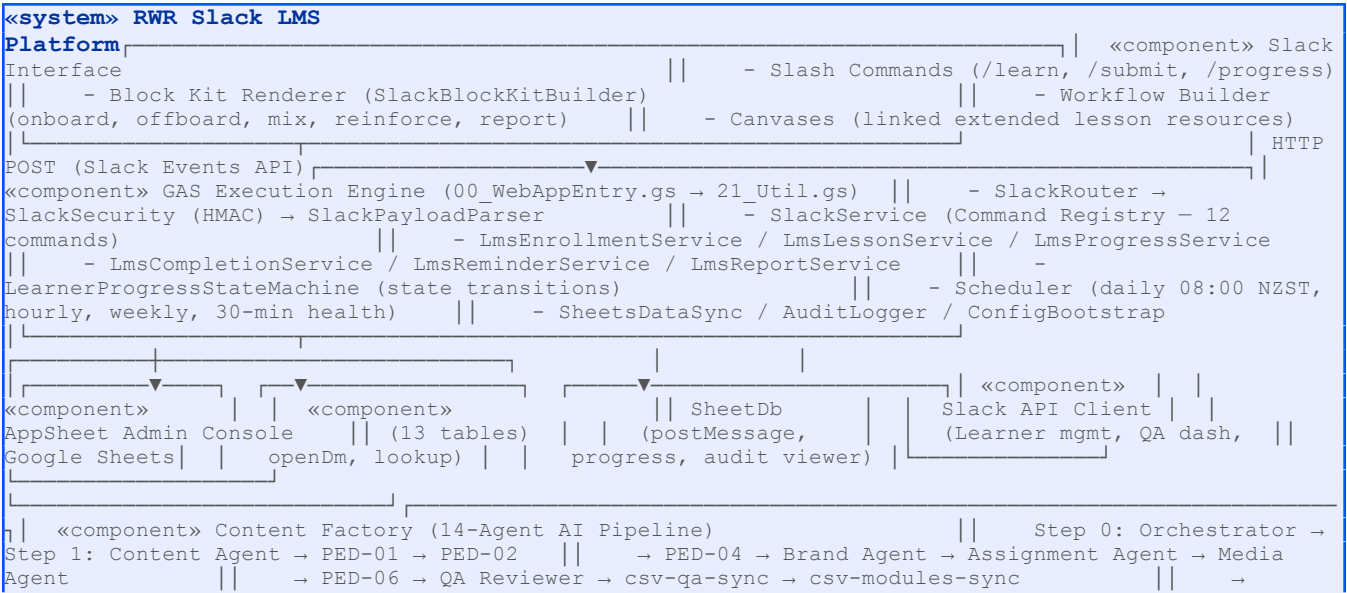
2.1 Platform Overview — 5-Layer Architecture

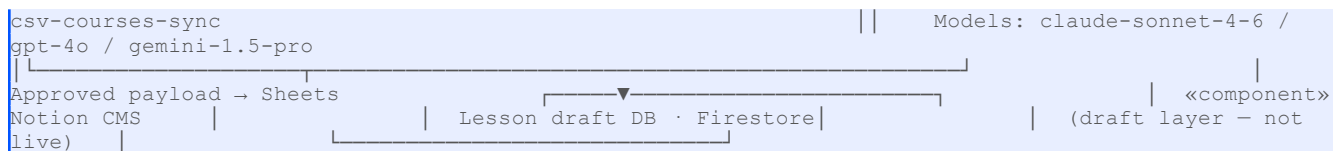
The RWR Slack LMS is a 5-layer platform. Each layer has a single responsibility. Data flows in one direction. Slack is the learner-facing interface only — it never stores state or makes decisions.

Layer	Name	Technology	Responsibility	Data Authority
L1	Interface	Slack API · Block Kit · Workflow Builder	Lesson delivery via DM · submission collection · admin workflow triggers · learner notifications	Read-only: renders content. Write: accepts /submit and admin commands.
L2	Execution Runtime	Google Apps Script (GAS V8) · doPost() · Scheduled Triggers	All slash command processing · lesson scheduling · SheetDb writes · Block Kit formatting · webhook handler for Workflow Builder	Sole writer to all system and pipeline columns in SheetDb. No direct Sheets access by any other layer.
L3	Data Store	Google Sheets (SheetDb v2+) · 13 tables · AppSheet (read-heavy)	Sole operational data store · all learner state · lesson content (approved) · progress · audit · QA records · module/course rollups	GAS is sole writer to system columns. AppSheet writes to user-input columns only. No other layer writes to Sheets.
L4	Content CMS	Notion (Notion MCP v2) · 288-lesson DB · Firebase Firestore (drafts)	Lesson draft authoring · QA history · curriculum structure · payload draft store (Firestore = draft layer only; Sheets = approved layer)	Content Operations writes. QA Reviewer reads and writes verdict. Notion is draft-authoring; Sheets is canonical approved state.
L5	AI / Intelligence	Google AI Studio App · Anthropic Claude API · OpenAI · Gemini · 14-agent pipeline	Content factory (14-agent pipeline) · LLM-agnostic switching · SOP-05 scoring · PED validation · brand compliance · QA scoring	AI agents write to Firestore draft layer and propose CSV rows; human operator approves before any Sheets write. AI never writes progression data.

2.2 UML Component Diagram — System Overview

INFO: The following diagram describes the component relationships in UML Component Diagram notation. Each box is a component; arrows show interface dependencies.





2.3 System Architecture Governing Rules

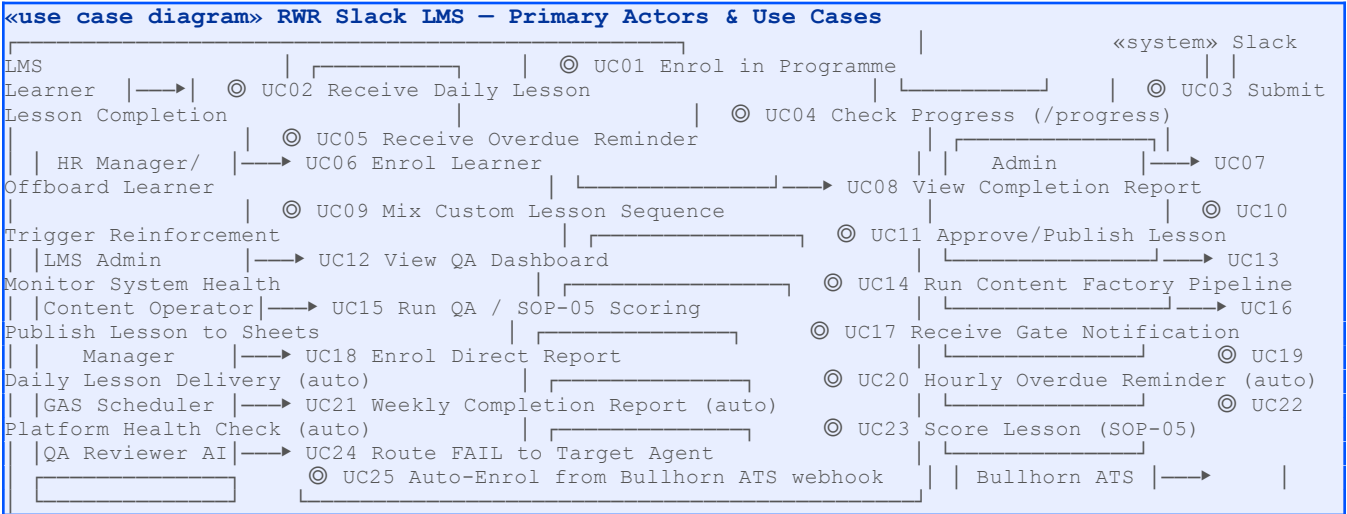
Rule #	Rule	Rationale
ARCH-01	Slack Never Waits: every action is a new, short GAS function. No long-running workflows.	Slack times out unresponsive webhooks at 3 seconds. Queue-first pattern (append to Sheets queue, return immediately) is mandatory.
ARCH-02	Database = Truth: all state lives in SheetDb (Google Sheets). Slack is stateless display only.	Slack messages are ephemeral. The database is the single source of truth for all learner state, progress, and lesson status.
ARCH-03	GAS is sole writer to system columns. AppSheet writes user-input columns only.	Prevents data corruption from concurrent writes. Column ownership is enforced at the AppSheet security rule level.
ARCH-04	Event-Driven Progression: submission → check. Click → check. Override → check. No timer-based progression.	Timers create race conditions. All progression is driven by explicit learner or admin events.
ARCH-05	Idempotent Operations: every click is safe if clicked twice. Database checks before acting.	Slack may deliver duplicate events. All write operations check for existing state before modifying.
ARCH-06	AI Never Writes Progression: AI reads and suggests only. The database decides.	AI agents propose lesson content and verdicts; human operators approve before any Sheets write.
ARCH-07	Two-Engine Separation: Content Generation Engine never sends to learners. Lesson Sending Engine never modifies content.	Prevents accidental publishing. The only shared surface is Google Sheets (Sheets is read-only for the Sending Engine).
ARCH-08	Skill-First: no workflow step exists without a corresponding Compendium skill.	Prevents ad-hoc code. Every automation is traceable to a named skill in the Skills Compendium.

3. Use Cases

3.1 Actor Definitions

Actor	Type	Description	Primary Use Cases
Learner	Primary Human Actor	A recruiting professional enrolled in the 12-month programme. Interacts exclusively via Slack DM.	UC01 (Enrol), UC02 (Receive Lesson), UC03 (Submit), UC04 (Check Progress), UC05 (Receive Reminder)
HR Manager / Admin	Primary Human Actor	Manages learner enrolment, monitors progress, generates reports. Uses Slack admin commands and AppSheet console.	UC06 (Enrol Learner), UC07 (Offboard), UC08 (View Report), UC09 (Mix Lessons), UC10 (Trigger Reinforce)
LMS Administrator	Primary Human Actor	Approves lesson content, manages QA dashboard, monitors system health.	UC11 (Approve Lesson), UC12 (View QA Dashboard), UC13 (Monitor Health)
Content Operator	Primary Human Actor	Authors lessons using the 14-agent content factory. Initiates content generation pipeline.	UC14 (Run Content Factory), UC15 (Run QA Pipeline), UC16 (Publish Lesson)
Manager	Supporting Human Actor	Receives gate-outcome notifications for direct reports. Can enrol their own team.	UC17 (Receive Gate Notification), UC18 (Enrol Direct Report)
GAS Scheduler	System Actor	Automated time-based trigger. Delivers daily lessons, hourly reminders, weekly reports, 30-min health checks.	UC19 (Daily Delivery), UC20 (Hourly Reminder), UC21 (Weekly Report), UC22 (Health Check)
QA Reviewer (AI)	System Actor	Final content gate. Issues binding SOP-05 verdicts. Routes failures to target agents.	UC23 (Score Lesson), UC24 (Route FAIL)
Slack Workflow Builder	System Actor	Handles admin workflows (onboard, offboard, mix, reinforce, report) via webhook to GAS.	UC06, UC07, UC09, UC10, UC21
Bullhorn ATS	External System	Source of auto-enrolment events via webhook on candidate placement.	UC25 (Auto-Enrol from HRIS)

3.2 Use Case Diagram — Primary System Boundaries



3.3 Use Case Specifications

UC01 — Enrol in Programme (Learner Self-Enrol)

Field	Detail
Use Case ID	UC01
Name	Enrol in Programme
Actor	Learner (primary) · HR Manager (alternative initiator)
Preconditions	Learner exists as a Slack workspace member. Admin has ADMIN_USER_IDS configured. Learner is not already enrolled (Learners table check).
Trigger	Learner sends /learn or admin sends /enroll {email} in Slack
Main Success Scenario	1. Learner or admin triggers enrolment command. 2. GAS verifies HMAC signature (04_SlackSecurity.gs). 3. SlackService routes to LmsEnrollmentService. 4. LmsEnrollmentService checks Learners table — no existing row. 5. Creates Learner row (Email, SlackUserID, DisplayName, Brand, Role, StartDate, pathway=standard). 6. Queries first lesson from Lessons table (M01-W01-L01, Status=Live). 7. Creates DeliveryQueue row (Status=queued, ScheduledFor=next delivery window). 8. Posts welcome DM via SlackApiClient.postMessage() with Block Kit welcome block. 9. Writes AuditLog entry. 10. Returns {ok:true, code:ENROLLED} to Slack.
Alternative Flows	A1 (Already enrolled): Step 4 finds existing Learner row → returns {ok:true, code:ALREADY_ENROLLED} — idempotent. A2 (Pathway specified): /enroll {email} senior → sets pathway=senior, triggers intake assessment delivery before first lesson.
Postconditions	Learner row exists in Learners table. DeliveryQueue row queued for M01-W01-L01. Welcome DM received by learner. AuditLog entry written.
Exception Flows	E1 (HMAC invalid): returns {ok:false, code:SIGNATURE_MISMATCH}. E2 (Config missing): throws ConfigError → doPost returns CONFIG_ERROR. E3 (Slack API failure): logs SLACK_API_ERROR to AuditLog; welcome DM not sent; Learner row still created.
Non-Functional	Response to Slack < 3s (queue-first pattern). AuditLog written synchronously before return.

UC02 — Receive Daily Lesson

Field	Detail
Use Case ID	UC02
Name	Receive Daily Lesson
Actor	GAS Scheduler (primary) · Learner via /learn (alternative)
Preconditions	Learner has Status=active in Learners table. DeliveryQueue has a row with Status=queued and ScheduledFor ≤ now. Lesson exists with Status=Live.
Trigger	Daily GAS trigger (08:00 NZST per learner timezone) OR learner sends /learn
Main Success Scenario	1. Scheduler trigger fires runDailyLessonDelivery(). 2. SheetsDataSync queries DeliveryQueue for Status=queued rows where ScheduledFor ≤ now. 3. For each row: fetch Lessons row by LessonId. 4. SlackBlockKitBuilder formats Block Kit lesson message (Hook, Core Content, Insight, Takeaway, Mission, Verification, Submit Block). 5. SlackApiClient.openDm() opens DM with learner Slack user. 6. SlackApiClient.postMessage() delivers Block Kit message. 7. LearnerProgressStateMachine transitions: queued → delivered. 8. Learner_Progress row updated (Status=delivered, DeliveredAt=now). 9. DeliveryQueue row updated (Status=delivered). 10. AuditLog entry written.
Alternative Flows	A1 (/learn command): learner triggers synchronously → same flow, returns within 5s via queue-first. A2 (Canvas linked): if Lessons.CanvasUrl is non-null, Block Kit includes "Open extended lesson →" button linking to Slack Canvas.
Postconditions	Learner receives Block Kit DM. Learner_Progress Status=delivered. AuditLog entry written.
Exception Flows	E1 (No lesson available): DeliveryQueue empty for learner → {ok:false, code:NO_LESSON_AVAILABLE}. E2 (Slack API error): logs SLACK_API_ERROR; lesson re-queued for next trigger run (AttemptCount++). E3 (Lesson not Live): lesson skipped; next available lesson fetched.
Non-Functional	/learn response < 5s. runDailyLessonDelivery() for full cohort < 5 min.

UC03 — Submit Lesson Completion

Field	Detail
Use Case ID	UC03
Name	Submit Lesson Completion
Actor	Learner
Preconditions	Learner has received lesson (Learner_Progress Status=delivered). Lesson has a valid SubmitCommand.
Trigger	Learner sends /submit {lessonId} complete OR reacts  to the lesson DM
Main Success Scenario	1. Slack delivers /submit event to GAS doPost(). 2. HMAC verified. 3. SlackPayloadParser extracts lessonId and "complete". 4. SlashService routes to LmsCompletionService. 5. Checks Learner_Progress — Status must be delivered or started (not completed). 6. LearnerProgressStateMachine transitions: delivered → submitted → completed. 7. Learner_Progress updated: Status=completed, CompletedAt=now, SubmittedAt=now. 8. Triggers csv sync chain: csv-qa-sync → csv-modules-sync → csv-courses-sync. 9. Posts confirmation DM to learner with next lesson preview. 10. Queues next lesson in DeliveryQueue. 11. If lesson is Gate Week (M03-W04, M06-W04, M12-W04): triggers gate assessment flow. 12. AuditLog entry written with correlationId.
Alternative Flows	A1 (React ): Slack sends reaction_added event → same completion flow. A2 (Already completed): Step 5 finds Status=completed → returns {ok:true, code:ALREADY_COMPLETED} — idempotent.
Postconditions	Learner_Progress Status=completed. Next lesson queued. Module stats rolled up via csv sync chain. Confirmation DM sent. AuditLog entry.
Exception Flows	E1 (Lesson not in progress): {ok:false, code:LESSON_NOT_IN_PROGRESS}. E2 (Sync chain failure): completion recorded; sync logged as failed; retry via RetryQueue. E3 (Gate assessment trigger fails): completion still recorded; gate failure logged to AuditLog.
Non-Functional	/submit end-to-end including sync chain < 30s.

UC04 — Check Progress

Field	Detail
Use Case ID	UC04
Name	Check Progress
Actor	Learner
Preconditions	Learner is enrolled (Learners table has active row).
Trigger	Learner sends /progress in Slack
Main Success Scenario	1. HMAC verified. 2. LmsProgressService queries Learner_Progress for LearnerEmail = caller. 3. Counts completed, delivered, queued lessons. 4. Calculates current month and week position. 5. SlackBlockKitBuilder formats progress summary Block Kit message. 6. Posts DM to learner. 7. AuditLog entry.
Postconditions	Learner receives progress summary DM showing: lessons completed, current month/week, lessons remaining, next lesson topic.
Non-Functional	/progress response < 3s.

UC06 — Enrol Learner (Admin)

Field	Detail
Use Case ID	UC06
Name	Enrol Learner (Admin-initiated)
Actor	HR Manager / Admin via Slack Workflow Builder or /enroll command
Preconditions	Caller has UserRole=admin in Learners table (admin guard enforced). Target email is a valid Slack workspace member.
Trigger	Admin submits Slack Workflow Builder onboarding form OR sends /enroll {email} [{pathway}]

Field	Detail
Main Success Scenario	1. Workflow Builder collects: email, brand, role, pathway (optional). 2. Webhook POSTs to GAS doPost(). 3. HMAC verified. 4. LmsEnrollmentService creates Learner row with all fields including manager_id (from enroller's learner_id if manager role). 5. Schedules M01-W01-L01 in DeliveryQueue. 6. Posts welcome DM to new learner. 7. If manager enrolled: posts notification to enroller. 8. AuditLog entry.
Alternative Flows	A1 (Non-standard pathway): intake assessment delivered to learner before first lesson. A2 (Bullhorn webhook): UC25 — auto-enrol via HRIS webhook on candidate placement.
Postconditions	Learner enrolled, first lesson queued, welcome DM sent, manager notified (if applicable).

UC07 — Offboard Learner

Field	Detail
Use Case ID	UC07
Name	Offboard Learner
Actor	HR Manager / Admin
Trigger	Admin submits offboarding form in Slack Workflow Builder
Main Success Scenario	1. Workflow Builder collects: learner email, reason. 2. Webhook to GAS. 3. LmsEnrollmentService sets Learner_rowStatus=archived. 4. Cancels all pending DeliveryQueue rows (Status=cancelled). 5. Records offboard event in Learner_Progress. 6. Posts confirmation DM to admin. 7. If learner has manager_id: posts notification to manager. 8. AuditLog entry (actor=admin).
Postconditions	Learner.Status=archived. No further lessons delivered. All queued deliveries cancelled. Audit trail complete.

UC14 — Run Content Factory Pipeline

Field	Detail
Use Case ID	UC14
Name	Run Content Factory Pipeline (Full 14-Agent)
Actor	Content Operator
Preconditions	CMC Final Output Package (Step 7) approved. Lesson brief prepared with: topic, objective, month, week, lesson#, type, difficulty, tone, previous 5 lesson topics.
Trigger	Content Operator sends "Run full pipeline" or individual agent commands in the AI Studio session
Main Success Scenario	1. Orchestrator (Step 0) validates inputs; injects topic brief; initialises pipeline state. 2. Content Agent (Step 1) generates ULC draft (7 sections + submit command) in JSON. 3. PED-01 (Step 2) checks cognitive load — single concept, ≤2 new terms, mission readability. PASS → continue. HARD FAIL → stop, route to Content Agent. 4. PED-02 (Step 3) checks real-world transfer — workplace action, observable evidence, theory-only detection. PASS → continue. HARD FAIL → PED-02 provides mission/verification rewrite. 5. PED-04 (Step 4) checks motivation/SDT — autonomy, competence, relatedness. SOFT FAIL → log flag, continue. 6. Brand Agent (Step 5) applies RWR brand rules — terminology, banned words, tone preset, emoji ≤ 2. 7. Assignment Agent (Step 6) validates mission design — feasibility, < 5 min, tools available, submission format. 8. Media Agent (Step 7) assesses visual needs — brief only, no block authority. 9. PED-06 (Step 8) validates assessment validity — objective alignment, gaming detection. 10. QA Reviewer (Step 9) scores against SOP-05 (6 criteria, weighted). STRONG_PASS/PASS/CONDITIONAL_PASS → trigger sync chain. SOFT/HARD FAIL → route to target agent. 11. csv-qa-sync (Step 10) writes QA fields to Lesson_QA_Records.csv + updates Lessons.csv Status. 12. csv-modules-sync (Step 11) rolls up module stats. 13. csv-courses-sync (Step 12) rolls up course stats.
FAIL Routing	Hard FAIL: stop pipeline. Show revision instruction. Max 2 retries per agent. 3rd failure → HUMAN_REVIEW_REQUIRED. Soft FAIL: log flag, carry to QA. Sync chain only runs on PASS/CONDITIONAL_PASS or better.
Postconditions	Lesson JSON produced. CSV row ready for Lessons.csv. Lesson_QA_Records.csv updated. Modules.csv and Courses.csv rolled up (on PASS). Notion page updated with QA metadata.

4. User Stories

4.1 Epic Map

Epic ID	Epic Name	Actor	Priority	Stories
EP01	Learner Enrolment & Onboarding	Learner, HR Manager	MUST	US01–US06
EP02	Daily Lesson Delivery	Learner, GAS Scheduler	MUST	US07–US12
EP03	Lesson Submission & Completion	Learner	MUST	US13–US17
EP04	Progress Tracking	Learner, Manager	MUST	US18–US21
EP05	Admin Learner Management	HR Manager, Admin	MUST	US22–US28
EP06	Content Factory & QA Pipeline	Content Operator, QA Reviewer AI	MUST	US29–US38
EP07	Certification Gates	Learner, System	SHOULD	US39–US42
EP08	Reinforcement & Mix	HR Manager, Learner	SHOULD	US43–US46
EP09	Reporting & Analytics	HR Manager, LMS Admin	MUST	US47–US51
EP10	System Health & Operations	System Admin, Platform Architect	MUST	US52–US55
EP11	Role-Based Learning Paths	HR Manager, Learner	COULD	US56–US59
EP12	Manager Capabilities	Manager	COULD	US60–US63
EP13	Privacy, Consent & Compliance	LMS Platform Lead	SHOULD	US64–US67
EP14	Offboarding & Course Retirement	HR Manager, Admin	MUST	US68–US70

4.2 User Stories — EP01: Learner Enrolment & Onboarding

Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priority
US01	New recruiting professional	receive a welcome Slack DM with overview of my 12-month learning journey when I am enrolled	I know what to expect and feel supported from Day 1	Welcome DM delivered < 30s of enrolment. DM includes: programme overview, first lesson preview, /learn command instruction, HR contact.	MUST
US02	Learner	see my first lesson delivered automatically on Day 1 without needing to ask for it	I have a frictionless start with no setup burden	Lesson delivered at 08:00 learner timezone on Day 1. Lesson is M01-W01-L01. Block Kit format includes all 7 ULC sections.	MUST
US03	Learner	have my Slack profile (name, brand, role) captured automatically at enrolment	I don't have to fill in forms to get started	Learners table populated: Email, SlackUserID, DisplayName, Brand, Role, StartDate within 5s of enrolment.	MUST

Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priority
US04	HR Manager	enrol a new team member without needing developer help, using a Slack Workflow Builder form	I can onboard people quickly and independently	Onboarding form available in Slack. Submission triggers enrolment. Learner receives welcome DM. Admin receives confirmation. No code change required.	MUST
US05	HR Manager	see a confirmation message in Slack when enrolment succeeds	I know the learner is set up correctly	Confirmation DM to admin includes: learner name, email, course, pathway, start date, first lesson scheduled time.	MUST
US06	Senior recruiter (3+ years)	be assessed before enrolment to start at the right curriculum level	I don't repeat content I already know	Intake assessment (10 MCQ) delivered via Slack Block Kit. Score $\geq 70\%$: pathway=senior, starts M04. Score $< 70\%$: pathway=standard, starts M01.	COULD

4.3 User Stories — EP02: Daily Lesson Delivery

Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priority
US07	Learner	receive one 5-minute micro-lesson in my Slack DM at the same time each working day	I can build a consistent learning habit	Daily lesson delivered 08:00 learner timezone ± 5 min. Each lesson follows ULC (7 sections). Total content ≤ 500 words.	MUST
US08	Learner	receive one 30-minute deep-dive lesson each Friday	I get deeper conceptual grounding once a week	Weekly Deep lesson delivered Friday. Content $\leq 1,200$ words. DEEP type flagged in Lessons table.	MUST
US09	Learner	use /learn any time to pull my next lesson on demand	I have flexibility if I miss a scheduled delivery	"/learn" returns next queued lesson within 5s. If no pending lesson: returns helpful message. Lesson not duplicated.	MUST
US10	Learner	see an extended resource Canvas linked from my lesson	I can go deeper when I want to without the lesson being overloaded	Block Kit lesson includes "Open extended lesson →" button if Lessons.CanvasUrl is non-null. Canvas opens in Slack.	SHOULD
US11	Learner	not receive the same lesson twice	My progress is tracked accurately across sessions	DeliveryQueue and Learner_Progress prevent duplicate delivery. Idempotent check on every delivery attempt.	MUST
US12	Content Operator	have all lessons quality-checked before they ever reach learners	No malformed or low-quality content is delivered	Lesson Status must = Live before it appears in DeliveryQueue. Live requires QA verdict $\geq$ CONDITIONAL_PASS + Publish Panel approval.	MUST

4.4 User Stories — EP03: Lesson Submission & Completion

Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priority
US13	Learner	submit a completed lesson using a pre-built command at the bottom of every lesson	Submission is frictionless — I don't have to remember a lesson code	Submit Block auto-appended to every lesson: `'/submit {lessonId} complete`. React  also works. Both paths update Learner_Progress.	MUST
US14	Learner	receive a confirmation DM after submitting	I know my completion was recorded	Confirmation DM delivered < 5s of submission. Includes: lesson title, completion time, next lesson preview.	MUST
US15	Learner	have my submission count once even if I click submit twice	Accidental double-submission doesn't create duplicate records	LmsCompletionService is idempotent — second submit returns ALREADY_COMPLETED; no duplicate Learner_Progress row created.	MUST
US16	Content Operator	have every lesson include a Verification question that cannot be answered without doing the Mission	Learners actually complete the task, not just tick a box	PED-06 Hard FAIL gates any Verification answerable without completing the Mission. Assignment Agent validates mission feasibility.	MUST
US17	Learner	have my submission evidence (text, screenshot, link) captured where required	My learning actions are documented for my manager	Learner_Progress.Submission Text stores submission evidence. Block Kit submission form guides learner to correct format.	SHOULD

4.5 User Stories — EP06: Content Factory & QA Pipeline

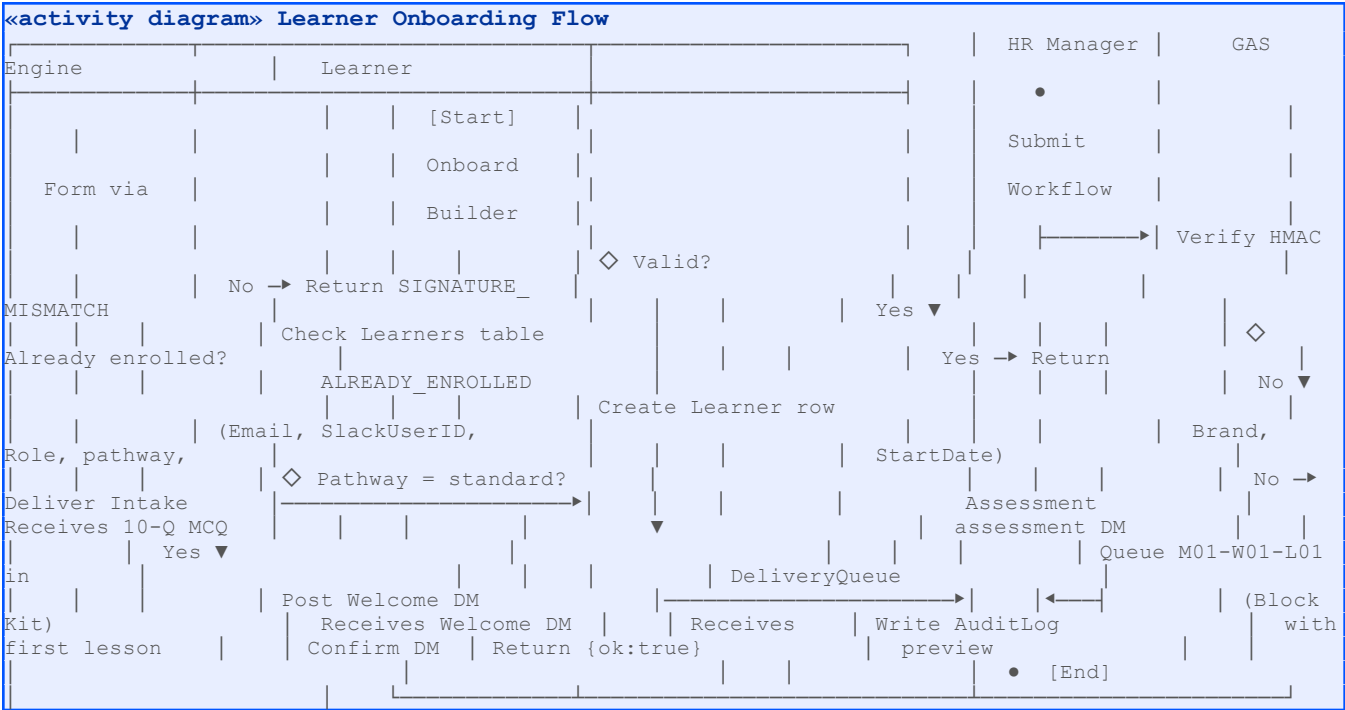
Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priority
US29	Content Operator	generate a lesson brief for a full week's lessons using the Content Map agent before any writing begins	I have a structured plan before content is generated	Content Map (Step 0) produces a brief for every slot in the week: topic, objective, tone, type, difficulty. No lesson text written at this step.	MUST
US30	Content Operator	run the 14-agent pipeline that validates each lesson against cognitive load, transfer, motivation, brand, mission, assessment, and QA criteria	Every lesson is practitioner-level and meets RWR's quality bar	Full pipeline (Steps 0–12) runs in sequence. Hard FAIL gates stop pipeline. Soft FAIL flags carried to QA. Only PASS+ lessons reach csv sync chain.	MUST
US31	Content Operator	have fabricated statistics automatically flagged and rejected during QA	No unverified claims reach learners	QA Reviewer Content Accuracy criterion (20%) scores 0 on any fabricated statistic. Insight section "directional language" enforced.	MUST
US32	QA Reviewer (AI)	route a failed lesson back to the exact target agent, not restart the full pipeline	Revision is efficient and targeted	FAIL output includes target_agent field. Orchestrator routes to target_agent only. Max 2 retries per agent. 3rd failure → HUMAN_REVIEW_REQUIRED.	MUST

Story ID	As a...	I want to...	So that...	Acceptance Criteria	Priori ty
US33	Content Operator	have the QA score and metadata automatically synced to Lessons.csv after QA completes	I don't have to manually update spreadsheets	csv-qa-sync (Step 10) writes QA_Status, SOP5_Score, QA_Verdict, QA_Detail to Lesson_QA_Records.csv on every PASS. Modules.csv and Courses.csv rolled up via Steps 11-12.	MUST
US34	Content Operator	have every Month 1 Week 1 lesson automatically flagged for human review regardless of QA score	Our most critical onboarding content has human eyes on it	Human Override Trigger: any M01-W01 lesson → QA automatically routes to HUMAN_REVIEW_REQUIRED regardless of SOP-05 score.	MUST

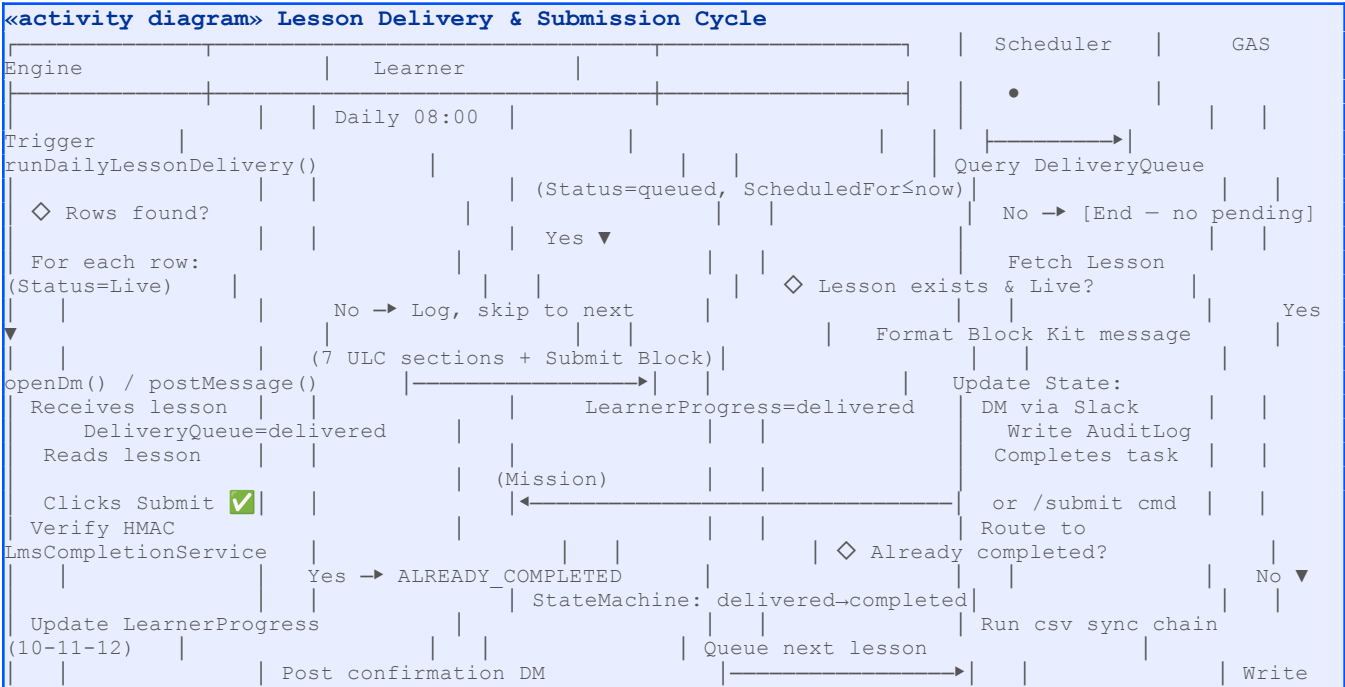
5. Activity Diagrams (UML)

INFO: Activity diagrams are expressed in UML Activity Diagram notation. Swim lanes show which actor or system component is responsible for each action. Decision nodes use diamond notation.

5.1 Activity Diagram — Learner Onboarding Flow

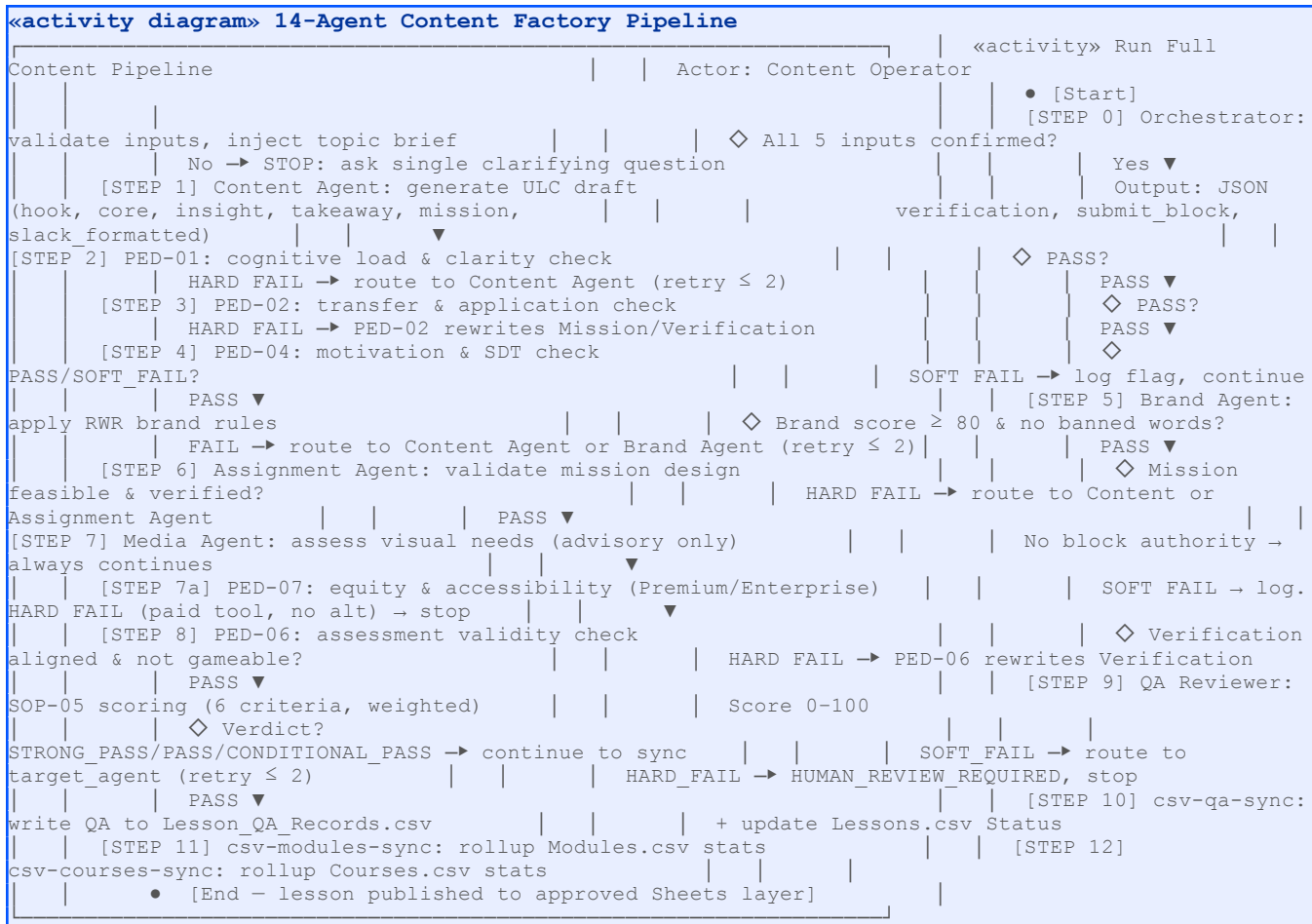


5.2 Activity Diagram — Lesson Delivery & Submission Cycle

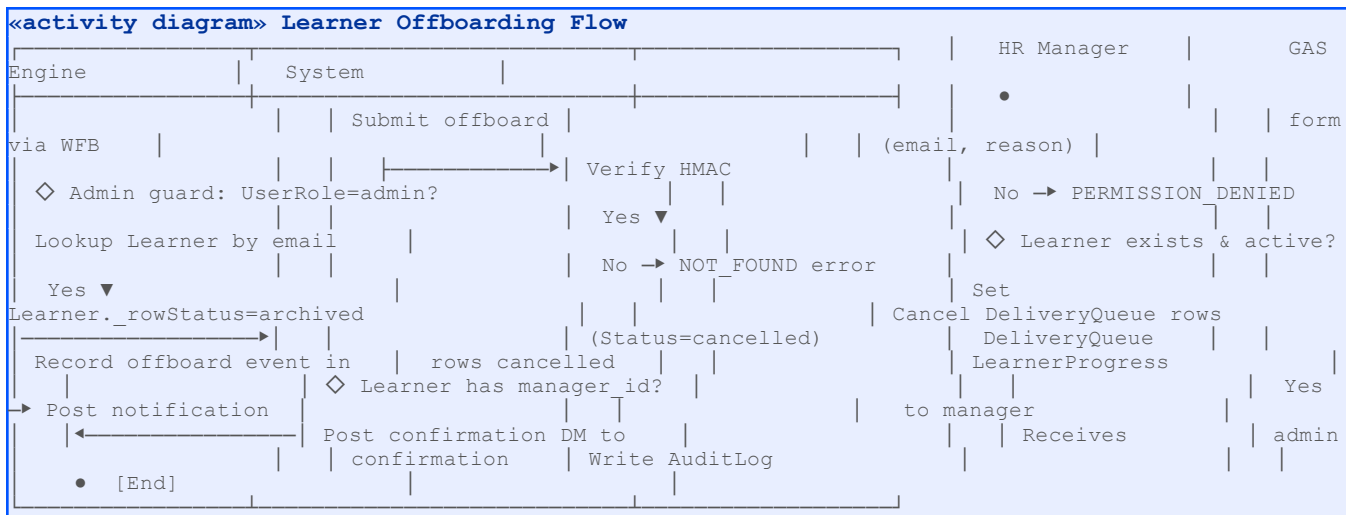




5.3 Activity Diagram — 14-Agent Content Factory Pipeline



5.4 Activity Diagram — Offboarding Flow



5.5 State Machine Diagram — Lesson Status Lifecycle

```

«state machine» Lesson Status Lifecycle (Empty → Live)
    •→ [Empty] | Trigger: Content author begins writing in Notion
    ▼ Actor: Content Operations [Draft] |
    Trigger: Author marks ready for QA / "Run full pipeline"
    ▼ Actor: Content Operator
    [SOP5Review] | Trigger: QA Reviewer scores ≥ CONDITIONAL_PASS | Alt: QA HARD_FAIL
    → [NeedsRevision] → [Draft] ▼ Actor: QA Reviewer (AI) [Ready] | Trigger:
    Operator approves Publish Panel | Precondition: testPassed=true, QA verdict ≥
    CONDITIONAL_PASS | csv-qa-sync writes to Lessons.csv ▼ Actor: Content Operator /
    LMS Administrator [Live] | Learner_Progress can now queue this lesson |
    DeliveryQueue entries valid | Alt: Incident rollback → [Ready] ▼ Actor: System
    (automatic on publish) •• [Terminal - in delivery]
  
```

5.6 State Machine Diagram — Learner Progress State

```

«state machine» LearnerProgress Status (queued → completed)
    •→ [queued] | Trigger: Lesson scheduled in DeliveryQueue
    | Actor: GAS Scheduler / LmsEnrollmentService ▼
    [delivered] | Trigger: Block Kit DM sent successfully to learner | Actor: GAS
    Scheduler / runDailyLessonDelivery() | Alt: DeliveredAt + 3 days with no submit →
    [overdue] ▼ [started] ← Optional intermediate state (learner opens lesson)
    | [submitted] | Trigger: /submit command or ☒ reaction | Actor:
    Learner ▼ [completed] | Trigger: LearnerProgressStateMachine validates
    transition | Actor: LmsCompletionService | Effect: csv sync chain triggered; next
    lesson queued ▼ •• [Terminal - lesson complete] [overdue] → Hourly reminder DM
    sent → Back to [delivered] [overdue] → Admin /gaps report flags this learner
  
```


#	Column Name	Group	Data Type	Nullable	GAS Write	AppSheet	Description
11	Hook	ULC Content	Text (≤2 sentences)	No	Yes	Read-only	Opening hook. Creates curiosity or challenges an assumption.
12	CoreContent	ULC Content	Text (≤300 words)	No	Yes	Read-only	Main lesson body. Practitioner-level knowledge. No filler.
13	Insight	ULC Content	Text (≤50 words, 1 sentence)	No	Yes	Read-only	Single specific, counterintuitive, or actionable insight.
14	Takeaway	ULC Content	Text (≤15 words, 1 sentence)	No	Yes	Read-only	Actionable takeaway. No vague platitudes.
15	Mission	ULC Content	Text	No	Yes	Read-only	Workplace action. Verb-first. Completable in <5 min using existing tools.
16	Verification	ULC Content	Text (1 question)	No	Yes	Read-only	Evidence question. Cannot be answered without completing the Mission.
17	SubmitBlock	ULC Content	Text	No	Yes	Read-only	Auto-generated submit block. Format: /submit {LessonID} complete.
18	SlackThreadText	Delivery	Text (full Slack-formatted lesson)	No	Yes	Read-only	Complete Block Kit-ready lesson text. Sent verbatim by Lesson Sending Engine.
19	SubmitCommand	Delivery	String (/submit M##-W##-L## complete)	No	Yes	Read-only	Pre-built submit command derived from LessonID. Auto-generated.
20	WC_Total	Metrics	Integer	Yes	Yes	Read-only	Total word count. Must be ≤500 (DailyMicro) or ≤1200 (WeeklyDeep).
21	WC_Core	Metrics	Integer	Yes	Yes	Read-only	Core Content word count. Must be ≤300.
22	WC_Insight	Metrics	Integer	Yes	Yes	Read-only	Insight word count. Must be ≤50.
23	WC_Takeaway	Metrics	Integer	Yes	Yes	Read-only	Takeaway word count. Must be ≤15.
24	Status	Lifecycle	String (Empty Draft SOP5Review NeedsRevision Ready Live)	No	Yes	Read-only	Lesson lifecycle status. Only Live lessons enter DeliveryQueue.
25	QA_Status	QA	String (Pending Passed Failed ConditionalPass HumanReview)	Yes	Yes	Read-only	QA workflow status. Written by csv-qa-sync.
26	SOP5_Score	QA	Integer (0–100)	Yes	Yes	Read-only	SOP-05 composite score from QA Reviewer.
27	QA_Verdict	QA	String (STRONG_PASS PASS CONDITIONAL_PASS SOFT_FAIL HARD_FAIL)	Yes	Yes	Read-only	QA Reviewer verdict. Written by csv-qa-sync.
28	QA_Detail	QA	Text (JSON)	Yes	Yes	Read-only	Full QA output: per-criterion scores, PED flags, target_agent, revision_prompt.

#	Column Name	Group	Data Type	Nulla ble	GAS Write	AppShe et	Description
29	QA_RunDate	QA	Date	Yes	Yes	Read-onl y	Date of most recent QA run.
30	BrandComplianc eScore	QA	Integer (0–100)	Yes	Yes	Read-onl y	Brand Agent compliance score. Must be ≥80 for publication.
31	PEDFlags	QA	Float (bitmask or JSON)	Yes	Yes	Read-onl y	Flags from PED agents (PED01_FAIL, PED02_REWRITE, etc.).
32	HumanReviewR equired	QA	Boolean	No	Yes	Read-onl y	True if M01-W01 lesson, 3rd FAIL attempt, or banned word detected.
33	MediaRequired	Content	Boolean	No	Yes	Read-onl y	True if Media Agent recommends visual support.
34	MediaBrief	Content	Float/Text (JSON or null)	Yes	Yes	Read-onl y	Media Asset Brief from Media Agent (asset_type, description, palette).
35	PublishedDate	Lifecycle	Date	Yes	Yes	Read-onl y	Date lesson reached Status=Live.
36	PublishedBy	Lifecycle	String (email)	Yes	Yes	Read-onl y	Operator email who approved publication.
37	MetadataVersion	System	Integer	No	Yes	Read-onl y	Schema version. Current: 1. Increment on breaking schema changes.
38	_createdAt	System	DateTime (ISO 8601 UTC)	No	Yes	Read-onl y	Row creation timestamp. GAS sets on first write.
39	_updatedAt	System	DateTime (ISO 8601 UTC)	No	Yes	Read-onl y	Last update timestamp. GAS updates on every write.
40	_createdBy	System	String (email or system)	No	Yes	Read-onl y	Actor who created the row.
41	_rowStatus	System	String (active archived)	No	Yes	Read-onl y	Soft-delete flag. Archived rows excluded from all queries.

6.3 Table: Learners

Column	Type	Nulla ble	Owner	Notes
Email	String (PK)	No	GAS	Slack user email. Primary key. Unique.
SlackUserID	String	No	GAS	Slack U... user ID. Used for openDm().
DisplayName	String	No	GAS+Ap pSheet	Display name from Slack profile. AppSheet can update.
Brand	String (RWRHealth Hospoworld Retailworld RWRConstruc tion)	No	GAS+Ap pSheet	Learner's specialist brand. AppSheet write-safe.
Role	String	No	GAS+Ap pSheet	Job role. e.g. "Junior Recruiter". AppSheet write-safe.
UserRole	String (learner admin)	No	GAS+Ap pSheet	System role. Controls admin command access.
StartDate	Date	No	GAS	Enrolment date. Set on first enrolment.
pathway	String (standard senior team_lead branch_manager)	No	GAS	Learning pathway. Determines curriculum start point.

Column	Type	Nulla ble	Owner	Notes
intake_score	Float (0.0–1.0)	Yes	GAS	Intake assessment score. Null if pathway=standard (no intake required).
intake_pathway_rec ommended	String	Yes	GAS	System-recommended pathway based on intake_score.
learner_id	String (UUID)	No	GAS	Internal learner ID. Referenced by Learner_Progress, manager_id.
manager_id	String (UUID FK → learner_id)	Yes	GAS+Ap pSheet	Self-referential FK. Points to manager's learner_id. Null if no manager.
client_id	String	Yes	GAS	Multi-tenant client identifier. Null for single-tenant deployments.
time_zone	String (IANA tz)	Yes	GAS+Ap pSheet	Learner timezone. e.g. "Pacific/Auckland". Used by Scheduler.
consent_given	Boolean	No	GAS	Privacy consent captured at enrolment. Required before first lesson.
consent_date	DateTime	Yes	GAS	Date/time consent was captured.
AdminNote	Text	Yes	AppShe et	Admin-only notes. AppSheet write-safe. Never GAS-written.
_createdAt	DateTime	No	GAS	Row creation timestamp.
_updatedAt	DateTime	No	GAS	Last update timestamp.
_createdBy	String	No	GAS	Actor who created the row.
_rowStatus	String (active archived)	No	GAS+Ap pSheet	Soft-delete. Archived = offboarded learner.

6.4 Table: Learner_Progress

Column	Type	Nulla ble	Notes
_Key	String (formula: LearnerEmail+" "+LessonID)	No	Composite primary key. Formula column — do not edit directly.
LearnerEmail	String (FK → Learners.Email)	No	Learner foreign key.
LessonID	String (FK → Lessons.LessonID)	No	Lesson foreign key.
CourseID	String (FK → Courses.CourseID)	No	Course foreign key. Denormalised for query performance.
Status	String (queued delivered started submitted completed overdue)	No	State machine state. Transitions enforced by LearnerProgressStateMachine.
DeliveredAt	DateTime	Yes	Set when Block Kit DM is confirmed sent.
StartedAt	DateTime	Yes	Optional. Set when learner opens lesson (if tracked).
SubmittedAt	DateTime	Yes	Set when /submit or  received.
CompletedAt	DateTime	Yes	Set when StateMachine transitions to completed.
DueAt	DateTime	Yes	Calculated due date for overdue detection.
SubmissionText	Text	Yes	Raw submission evidence from learner (text, URL, or description).
AttemptCount	Integer	No	Number of delivery attempts. Default 1.

Column	Type	Nulla ble	Notes
_createdAt	DateTime	No	Row creation timestamp.
_updatedAt	DateTime	No	Last update timestamp.
_createdBy	String	No	Actor: system / scheduler / learner email.

6.5 Table: Modules

Column	Type	Nulla ble	Notes
ModuleID	String (M## — PK)	No	Primary key. e.g. M01, M02 ... M12.
CourseID	String (FK → Courses.CourseID)	No	Parent course foreign key.
Title	String	No	Module title. e.g. "Month 1: Foundations of Recruitment".
DifficultyTier	String (Guided Independent Strategic)	No	Tier for this module.
TotalLessons	Integer	No	Total lessons in module (always 24 for standard months).
PublishedLessons	Integer	Yes	Count of lessons with Status=Live. Written by csv-modules-sync.
AvgQAScore	Float	Yes	Average SOP-05 score across all Live lessons. Written by csv-modules-sync.
QAPassRate	Float (0.0–1.0)	Yes	Proportion of lessons with QA_Verdict ≥ CONDITIONAL_PASS.
Status	String (Empty Outlined Writing Review Ready)	No	Module production status.
GateWeek	String or null	Yes	Gate week reference. e.g. "M03-W04" or null.
LastUpdated	DateTime	Yes	Last csv-modules-sync write.
_createdAt	DateTime	No	Row creation timestamp.
_updatedAt	DateTime	No	Last update timestamp.
_rowStatus	String (active archived)	No	Soft-delete flag.

6.6 Table: DeliveryQueue

Column	Type	Nulla ble	Notes
QueueId	String (UUID — PK)	No	GAS-generated UUID. Primary key.
LearnerEmail	String (FK → Learners.Email)	No	Learner foreign key. NOT NULL — resolved before insertion.
LessonId	String (FK → Lessons.LessonID)	No	Lesson foreign key. NOT NULL — lesson must exist and be Live before insertion.
EnrollmentId	String	Yes	FK to enrollment event for traceability.
ScheduledFor	DateTime	No	Target delivery time in learner's local timezone.
Status	String (queued delivered failed cancelled)	No	Queue item status. Scheduler processes queued rows only.
AttemptCount	Integer	No	Delivery attempt count. Default 0. Incremented on each failed delivery.
CreatedAt	DateTime	No	Row creation timestamp.

Column	Type	Nulla ble	Notes
UpdatedAt	DateTime	No	Last update timestamp.
DeletedAt	DateTime	Yes	Soft-delete timestamp (used for cancelled items).

6.7 Table: AuditLog (Append-Only — No Updates, No Deletes)

 **RULE:** AuditLog is immutable. GAS must NEVER update or soft-delete audit rows. No _rowStatus or UpdatedAt columns. Audit records are the legal and compliance trail for all system operations.

Column	Type	Nulla ble	Notes
AuditId	String (UUID — PK)	No	GAS-generated UUID. Primary key.
Timestamp	DateTime (ISO 8601 UTC)	No	Event timestamp at GAS execution time.
Actor	String	No	Who performed the action: system scheduler {learner_email} {admin_email}.
Action	String	No	What happened: LESSON_DELIVERED, LESSON_COMPLETED, LEARNER_ENROLLED, etc.
ResourceType	String	No	Entity type: Lesson Learner Module Course System.
ResourceId	String	No	Primary key of the affected entity.
CorrelationId	String (UUID)	No	Links related operations across tables. Same correlationId for submit + sync chain entries.
Status	String (SUCCESS ERROR WARN)	No	Operation outcome.
Message	Text	Yes	Human-readable description. Includes error codes on ERROR status.
Metadata	Text (JSON string)	Yes	Additional context: command arguments, lesson topic, error detail, etc.
Source	String (slash_command scheduler webhook admin system)	No	Event source.

6.8 Additional Tables — Schema Summary

Table	PK	Key Columns	Notes
Courses	CourseID (COURSE_[NAME])	Title, Modules (pipe-sep), TotalLessons, PublishedLessons, CompletionRate, AvgQAScore, Status	Master course table. 1 row per deployable course. Updated by csv-courses-sync.
Lesson_QA_Records	LessonID	QA_Status, SOP5_Score, QA_Verdict, QA_Detail, RunDate, BrandComplianceScore, PEDFlags, HumanReviewFlag	QA output table. Written by csv-qa-sync (Step 10) on every PASS+ verdict. Always-overwrite (not empty-only).
Lesson_Metrics	LessonID	WC_Total, WC_Core, WC_Insight, WC_Takeaway, BrandComplianceScore, PEDFlags	Word count and metric data. Written by Content Agent output. Updated by Brand Agent and PED agents.
Assessments	AssessmentId (UUID)	LearnerEmail FK, LessonId FK, Type (GateWeek IntakeAssessment), Score, Passed, AttemptCount, PassTimestamp, EvidenceUrl	Gate Week and Intake Assessment results. Append-only per attempt. Separate from Learner_Progress completion.

Table	PK	Key Columns	Notes
RetryQueue	RetryId (UUID)	JobType, Payload (JSON), Attempts (Integer), NextRunAt, Status (queued dead_letter), LastError	Dead-letter queue for failed operations. processDLQ() runs on GAS scheduled trigger with exponential backoff.
Schedules	id (UUID)	enrollmentId, studentId, lessonId, scheduledFor, status, attemptCount	Delivery schedule. Replaces old delivery_queue. LessonId NOT NULL.
Pathway_Map	PathwayId	pathway, month_start, lesson_sequence (JSON array)	Role-based pathway to lesson mapping. Phase 2 build. Required before non-standard pathway delivery.

7. Data Dictionary

7.1 Enumerated Field Values

7.1.1 Lesson Status Lifecycle

Value	Meaning	Actor Who Sets	Transition From
Empty	No content exists for this lesson slot	Content Operations (creates Notion page)	— (initial state)
Draft	Content author has begun writing	Content Operations	Empty
SOP5Review	Ready for QA pipeline run	Content Operator (triggers pipeline)	Draft
NeedsRevision	QA pipeline returned SOFT_FAIL or HARD_FAIL	QA Reviewer (AI)	SOP5Review
Ready	QA passed; Publish Panel approved; awaiting deployment	Content Operator (Publish Panel)	SOP5Review
Live	Lesson deployed; appears in DeliveryQueue for learners	GAS (csv-qa-sync writes publishedDate)	Ready

7.1.2 QA Verdict Values

Value	Score Range	Publication Action	Sync Chain
STRONG_PASS	90–100	Auto-approve. Flag as Golden Example candidate.	Run Steps 10–12
PASS	80–89	Auto-approve.	Run Steps 10–12
CONDITIONAL_PASS	60–79	Approve + flag for human spot-check.	Run Steps 10–12
SOFT_FAIL	40–59	Do not publish. Route to target_agent (max 2 retries).	DO NOT run — stays Draft
HARD_FAIL	0–39	Immediate human review required. Pipeline stops.	DO NOT run — stays Draft

7.1.3 Learning Pathway Values

Pathway ID	Target Role	Curriculum Start	Intake Required	Gate Score
standard	Junior Recruiter / New Starter	M01-W01-L01 (full 12 months)	No	N/A
senior	Experienced Consultant (3+ years)	M04-W01-L01 (skips M01–M03)	Yes (10 MCQ)	≥70%
team_lead	Team Lead / Senior Consultant	M07-W01-L01 (includes M09–M12 strategic)	Yes (10 MCQ)	≥80%
branch_manager	Branch Manager	M09-W01-L01 (strategic tier only, ~6 months)	Yes (10 MCQ)	≥80% + manager override

7.1.4 Lesson Intent Values (6-Intent Taxonomy)

Intent	Meaning	Mission Pattern	Example
Orient	Show me the map. Where am I? What's the big picture?	Explore, map, or overview task	Here's how onboarding works and what to expect
Explain	Teach me the concept. How does this work?	Read, watch, or analyse and summarise	Here's how our CRM system tracks candidates
Practise	Let me try it. Build the skill through repetition.	Do the task 3–5 times using real tools	Write 5 cold emails using this template
Decide	Help me choose. What do I do in this scenario?	Review a case, choose a path, justify	Should you move this candidate forward? Review the brief.
Reflect	Let me think back. What did I learn? What's next?	Journal, retrospective, or peer sharing	Write 3 takeaways from your first month
Comply	I must follow this rule. It's mandatory.	Complete the required action and confirm	Complete this GDPR awareness exercise to continue

7.2 Key Business Terms Glossary

Term	Definition	System Context
ULC (Universal Lesson Canvas)	The mandatory 7-section structure for every lesson: Hook → Core Content → Insight → Takeaway → Mission → Verification → Submit Block.	Enforced at QA gate (SOP-05 criterion 1). Missing any section = -15 pts. Missing Submit Block = 0 on criterion.
SOP-05	The 6-criterion weighted QA rubric: ULC Completeness (20%), Word Limits (15%), Brand Compliance (15%), Mission Feasibility (20%), Content Accuracy (20%), Continuity (10%).	Applied by QA Reviewer (AI) at Step 9. Score 0–100. Thresholds: 90=STRONG_PASS, 80=PASS, 60=CONDITIONAL_PASS, 40=SOFT_FAIL, 0=HARD_FAIL.
Skills Compendium	The registry of 77 named skills (March 2026) that are the building blocks of every automation workflow. No workflow step may exist without a Compendium skill reference.	Referenced at WAA Step S1. Skill identifiers are used in GAS file manifest and pipeline skill mapping tables.
CMC (Content Mapping Controller)	The mandatory 7-step pre-pipeline workflow (RWR-LMS-SOP-001). Must complete before any Content Agent run.	CMC Steps 0–7. Step 7 Final Output Package requires operator sign-off. Content Agent cannot run before Step 7 approval.
SheetDb	The Google Sheets-backed database client library (v2+). Provides a table/schema abstraction over Sheets tabs.	GAS uses SheetDb for all database reads and writes. Schema registered via <code>db.schema()</code> . CRUD via <code>db.table().find()</code> , <code>.create()</code> , <code>.update()</code> , <code>.remove()</code> .
WAA (Workflow Automation Architect)	The 9-step design process (S0–S8) that every automation workflow must complete before any code is written.	Mandatory for all new workflows. Two approval checkpoints: Checkpoint 1 (Design Approval) after S2; Checkpoint 2 (Implementation Sign-off) after S8.
Block Kit	Slack's structured message format using JSON-defined UI blocks. Used for all lesson delivery and interactive elements.	<code>SlackBlockKitBuilder</code> (12_SlackBlockKitBuilder.gs) generates all Block Kit JSON. Never use plain text for lesson delivery.
DeliveryQueue	The scheduling table that queues lessons for delivery to learners. LessonId FK is NOT NULL — lesson must be Live before insertion.	Populated by <code>LmsEnrollmentService</code> on enrol and <code>LmsCompletionService</code> on lesson completion. Processed by <code>runDailyLessonDelivery()</code> .
correlationId	A UUID that links related operations across the AuditLog. e.g., /submit event + csv-qa-sync write + csv-modules-sync write all share the same correlationId.	Generated by GAS at the start of each multi-step operation. Enables full audit trail reconstruction for compliance.

Term	Definition	System Context
PED-07 (Equity & Accessibility)	The optional UDL-based accessibility agent. Active for Premium/Enterprise deployments. Checks plain language, cultural assumptions, tool dependency, and cognitive accessibility.	Soft FAIL (log). Hard FAIL only when mission requires a paid tool with no alternative path. Disabled by default at Standard tier.
Gate Week	A milestone week in the curriculum that triggers a certification assessment: M03-W04 (90-Day Gate), M06-W04 (Mid-Programme Gate), M12-W04 (Graduation).	Lesson tone switches to tone-06 (Cert-Formal) or tone-07 (Celebratory). Completion triggers Assessments table entry and manager notification.
Banned Words	Eight words whose presence in any final lesson output triggers automatic HARD_FAIL and human review, regardless of SOP-05 score.	List: "just a recruiter", "human resources", "staff", "manpower", "headcount", "leverage", "synergy", "transformative".

8. CSV File Specifications

8.1 Lessons.csv — 41-Column Canonical Schema

i INFO: Lessons.csv is the approved-payload surface and the contract between the Content Factory (Engine 1) and the Lesson Sending Engine (Engine 2). Column order is mandatory and must not change without incrementing MetadataVersion.

Col #	Column Name	Data Type	Example Value	Validation Rule
1	LessonID	String	M03-W02-L04	Format: M###-W###-L## or M###-W###-DEEP. Must match Modules table.
2	ModuleID	String	M03	Format: M### (01–12). FK to Modules.ModuleID.
3	Week	Integer	2	Range: 1–4.
4	LessonSlot	String	L04	Values: L01, L02, L03, L04, L05, DEEP.
5	LessonType	String	DailyMicro	Values: DailyMicro WeeklyDeep.
6	Topic	String	Boolean Search for Niche Technical Roles	Non-empty. Max 200 chars.
7	Objective	String	Apply boolean operators to reduce results by 80%	Non-empty. Must be verb-first.
8	Intent	String	Practise	Values: Orient Explain Practise Decide Reflect Comply.
9	DifficultyTier	String	Independent	Values: Guided Independent Strategic.
10	TonePreset	String	tone-01	Values: tone-01 through tone-07.
11	Hook	String	Most boolean searches return 10x too many results. Here's why.	Max 2 sentences. Non-empty.
12	CoreContent	String	[lesson body text]	Max 300 words.
13	Insight	String	Using NOT eliminates 60% of noise faster than AND stacking.	Max 50 words. 1 sentence.
14	Takeaway	String	Start every search with one exclusion, then refine.	Max 15 words.
15	Mission	String	Open LinkedIn Recruiter and build one boolean string...	Verb-first. Completable <5 min.
16	Verification	String	Paste the boolean string you built. How many results did it return?	1 question. Cannot be answered without completing Mission.
17	SubmitBlock	String	👉 *Done? Submit it:* `/submit M03-W02-L04 complete`	Auto-generated. Must contain /submit {LessonID} complete.
18	SlackThreadText	String	[full Slack-formatted lesson]	Non-empty. Slack mrkdwn format.
19	SubmitCommand	String	/submit M03-W02-L04 complete	Format: /submit {LessonID} complete.
20	WC_Total	Integer	312	≤500 for DailyMicro; ≤1200 for WeeklyDeep; ≥150 always.

Col #	Column Name	Data Type	Example Value	Validation Rule
21	WC_Core	Integer	245	≤300 always.
22	WC_Insight	Integer	22	≤50 always.
23	WC_Takeaway	Integer	9	≤15 always.
24	Status	String	Ready	Values: Empty Draft SOP5Review NeedsRevision Ready Live.
25	QA_Status	String	Passed	Values: Pending Passed Failed ConditionalPass HumanReview.
26	SOP5_Score	Integer	87	Range: 0–100. Null until QA runs.
27	QA_Verdict	String	PASS	Values: STRONG_PASS PASS CONDITIONAL_PASS SOFT_FAIL HARD_FAIL.
28	QA_Detail	String (JSON)	{"ULC":20,"WordLimits":15,"Brand":14,...}	JSON string. Min keys: per-criterion scores.
29	QA_RunDate	String (date)	2026-04-04	ISO 8601 date. YYYY-MM-DD.
30	BrandCompliance Score	Integer	92	Range: 0–100. ≥80 required for publication.
31	PEDFlags	String	PED04_SOFT_FAIL	Comma-separated flag codes. Empty if no flags.
32	HumanReviewRequired	Boolean	FALSE	TRUE FALSE. Auto-set for M01-W01 and 3rd FAIL.
33	MediaRequired	Boolean	FALSE	TRUE FALSE.
34	MediaBrief	String (JSON or empty)		JSON if MediaRequired=TRUE, empty otherwise.
35	PublishedDate	String (date)	2026-04-10	ISO 8601 date. Set by csv-qa-sync on Status=Live.
36	PublishedBy	String (email)	ops@rwrgroup.com	Operator email. Set by Publish Panel.
37	MetadataVersion	Integer	1	Schema version. Current: 1.
38	_createdAt	String (datetime)	2026-04-04T09:00:00Z	ISO 8601 UTC. GAS auto-set on row creation.
39	_updatedAt	String (datetime)	2026-04-10T14:30:00Z	ISO 8601 UTC. GAS auto-set on every write.
40	_createdBy	String	system	system {operator email}.
41	_rowStatus	String	active	Values: active archived.

8.2 12-Month Employee Pathway CSV

Property	Value
File	12_Month_Employee_Pathway_SLACK.csv
Rows	1,777 rows (full Slack delivery workflow)
Key Columns	Category, Resource, Link, Quiz
Purpose	Maps all 288 lessons to Slack delivery triggers, message templates, and channel assignments. Used by Slack Workflow Builder for scheduled delivery sequencing.
Format	Comma-delimited, UTF-8, header row

8.3 Onboarding Checklist CSV

Property	Value
File	Onboarding_Checklist_workflow.csv
Rows	70 items
Columns	Category, Field/Question, Who Completes?
Purpose	70-item onboarding checklist across 3 completion roles: new hire, IT, HR. Used by Slack Workflow Builder onboarding form.
Who Completes values	New Hire IT HR Manager LMS Admin

9. Skills Compendium — Summary Register

i INFO: The Skills Compendium (March 2026, 77 skills) is the authoritative registry of all automation building blocks. No workflow step may exist without a corresponding Compendium skill. This section lists all skills by category with their identifier, purpose, and pipeline position.

9.1 Planning Skills

Identifier	Skill Name	Purpose in RWR LMS	Input	Output
content-map	Content Map	Decides topic, objective, tone, type, and difficulty for every lesson in a week before any writing begins. Turns a 35-question week into a 7-question week.	Month#, Week#, course context, previous 5 topics	Lesson brief per slot (topic, objective, tone, type, difficulty)
workflow-mapper	Workflow Mapper	Maps any agent workflow, automation, or business process to a file tree structure identifying instructions, tools, and data at each node.	Business process description, trigger, end state	File tree structure with instructions, tools, data at each node
data-mapper	Data Mapper	Maps what each workflow task reads and writes, where data lives, and what format it is in.	Workflow task structure	Per-task data map with source, transformation, output, storage
tool-mapper	Tool Mapper	Maps tool layer of agent workflows — which MCP servers, what permissions, what fallbacks.	Task structure, data sources	Tool registry per task with MCP server, permissions, fallback
execution-chain-writer	Execution Chain Writer	Designs the relay chain diagram and handoff table for a workflow.	Skill mapping table (S4)	Relay chain diagram + Skill Input Output On Failure table
orchestrator-executor	Orchestrator Executor	Manages pipeline state, routes failures, enforces max 2 retries per agent.	Pipeline state, current step, fail counts	Routing instruction + updated pipeline state

9.2 Content Generation Skills (14-Agent Pipeline)

Identifier	Step	Authority	Purpose	Hard/Soft FAIL
content-agent	Step 1	Creator	Generates complete ULC lesson draft (7 sections) from topic brief. Produces JSON + CSV row + Metrics row + REGISTER block + /submit command.	N/A — target of Hard FAILs from downstream
ped-01-cognitive-load	Step 2	Hard FAIL gate	Validates single concept, ≤2 new terms, mission readability. Checks cognitive budget (5 min/1 concept for DailyMicro).	Hard FAIL → route to Content Agent (rewrite)
ped-02-transfer	Step 3	Hard FAIL + Rewrite	Validates real-world workplace action, observable evidence, theory-only detection. Can rewrite Mission and Verification directly (authoritative).	Hard FAIL → PED-02 rewrite (authoritative) or Content Agent if full regen needed
ped-04-motivation	Step 4	Soft FAIL (log)	Checks autonomy, competence, relatedness. Flags controlling language	Soft FAIL → log, continue pipeline, carry to QA

Identifier	Step	Authority	Purpose	Hard/Soft FAIL
			("You must..." and hype language ("Let's crush it!").	
brand-agent	Step 5	Brand gate	Applies RWR brand rules: terminology, 8 banned words (zero tolerance), tone preset, emoji ≤ 2. Score 0–100.	Brand FAIL (score<80 or banned word) → Hard FAIL → route to Content or Brand Agent
assignment-agent	Step 6	Validator	Validates mission: <5 min, available tools, submission format, alternative path (if paid tool), verification integrity.	Hard FAIL → route to Content Agent or Assignment Agent
media-agent	Step 7	Advisory only	Assesses visual support needs. Produces asset brief (asset_type, description, palette). Default is no media needed.	No block authority — always continues
ped-07-accessibility	Step 7a	Soft/Hard FAIL	Checks plain language, cultural assumptions (NZ/AU), tool dependency, cognitive accessibility. Premium/Enterprise only.	Soft FAIL (log). Hard FAIL only when paid tool + no alt path.
ped-06-assessment	Step 8	Hard FAIL + Rewrite	Validates objective alignment, evidence quality, gaming detection, signal clarity. Can rewrite Verification directly (authoritative).	Hard FAIL → PED-06 rewrites Verification (authoritative)
qa-reviewer	Step 9	Final gate (Publication)	Scores against SOP-05 (6 weighted criteria). Issues binding verdict. Routes FAILs to target_agent. Max 2 retries per agent.	SOFT_FAIL → target_agent. HARD_FAIL → HUMAN_REVIEW_REQUIRED
csv-qa-sync	Step 10	Operations	Writes QA fields to Lesson_QA_Records.csv + updates Lessons.csv Status. Always-overwrite rule (not empty-only).	ABORT_LESSON_NOT_FOUND ABORT_MISSING_QA_DETAIL
csv-modules-sync	Step 11	Operations	Rolls up module stats in Modules.csv: PublishedLessons, AvgQAScore, QAPassRate, Status, LastUpdated.	Log discrepancy, continue
csv-courses-sync	Step 12	Operations	Rolls up course stats in Courses.csv: CompletionRate, AvgQAScore, TotalLessons, PublishedLessons, Status.	ABORT_INVALID_COURSE_ID
ped-03-retention	Curriculum-level	Soft FAIL	Checks retrieval practice, spacing (5–15 lessons between reinforcements), illusion-of-mastery detection. Never runs on individual lessons.	Soft FAIL — logs spacing issues, flags illusion lessons, recommends retrieval opportunities
ped-05-sequencing	Curriculum-level	Hard/Soft FAIL	Checks prerequisite coverage, scaffolding progression, week coherence, month arc. Never runs on individual lessons.	Hard FAIL (prerequisite violations). Soft FAIL (sequencing suggestions).

9.3 Data & Integration Skills

Identifier	Purpose
sheets-schema-writer-updated	Generates GAS SheetDb schema registration code for all tables. Must run after appsheet-data-modeler for any AppSheet-consumed table.
sheets-data-sync	Reads approved lessons from authoring Sheets; upserts to delivery runtime SheetDb. Bridge between content authoring and delivery.
appsheet-data-modeler	Produces AppSheet-compatible data model with key columns, ref columns, formula column flags, write-safe columns, and GAS-owned columns.
appsheet-app-writer	Generates AppSheet app configuration specification: table bindings, view list, navigation, brand colours, locale settings.
appsheet-security-rule-writer	Generates row security filters and column update rules for AppSheet. Named expressions: IsAdmin, IsLearner, MyBrand, MyEmail.
slack-api-caller	UrlFetchApp wrappers for Slack Web API: postMessage, openDm, updateMessage, lookupByEmail. Used by SlackApiClient.gs.
slack-block-kit-writer	Generates Block Kit JSON for lesson delivery, progress summaries, quiz blocks, assessment blocks, admin notifications.
slash-command-registry	Dispatches slash commands to correct service handlers. Enforces admin guard. Validates command arguments.
airtable-flexible-writer	Airtable write operations for any table/field combination. Used for external reporting integrations.
gas-service-writer	Generates GAS service class implementations for domain logic (LmsLessonService, LmsCompletionService, etc.).
gas-webapp-writer	Generates doPost() entry point, router, and DI factory for GAS web app deployments.
state-machine-writer	Generates state machine class with explicit valid transitions. Used by LearnerProgressStateMachine.gs.

9.4 Validation / QA Skills

Identifier	Purpose
qa-reviewer	Final SOP-05 scoring and verdict. Issues binding publication decisions. Routes FAILs to target_agent with revision_prompt.
retry-resolver	DLQ tab processing with exponential backoff. processDLQ() runs on GAS scheduled trigger.
vuln-scanner	Security vulnerability scanning for GAS codebase. Checks HMAC implementation, timestamp replay, admin guards, config exposure.
health-monitor	runHealthCheck() — checks Sheets connectivity, GAS quota, Slack API reachability every 30 min.
test-harness-writer	Generates 19_HostTestHarness.gs with unit tests, failure path tests, security tests, integration tests.
audit-logger	Immutable append-only AuditLog writer. All system operations must call db.audit() before returning.
config-bootstrapper	18_ConfigBootstrap.gs — validates all required Script Properties on doPost(). Throws ConfigError if any missing.
secret-rotation-writer	Generates zero-downtime secret rotation procedures for all 6 system secrets (SLACK_BOT_TOKEN, SLACK_SIGNING_SECRET, etc.).

 **NOTE:** csv-courses-sync and ped-07-accessibility are operationally specified in the WAA Technical Document (§5.7, §5.8) but are NOT yet registered in the Skills Compendium (March 2026 — 77 skills). Both must be registered before they can be invoked via the Orchestrator. Action: generate Skill Creation prompts (CMC Step 5 template) for both skills before the next pipeline deployment.

10. Workflow Designs

10.1 Workflow A — Content Factory: Lesson Generation

Step #	Skill	Input	Output	Gate	Storage
0	content-map	Month#, Week#, course context, 5 previous topics	Lesson brief per slot (topic, objective, tone, type, difficulty, M#-W#-L# ID)	Operator approves brief before Step 1	Memory (brief handed to Content Agent)
1	content-agent	Lesson brief from Step 0	JSON: hook, core, insight, takeaway, mission, verification, submit_block, slack_formatted, CSV row, Metrics row, REGISTER block	Creator — no gate authority	Firestore /payloads/{lessonId}/draft (In Progress)
2	ped-01-cognitive-load	Full lesson JSON from Step 1	PASS/FAIL verdict + issues list + revision_instruction	Hard FAIL → route to Content Agent (retry ≤ 2)	pipelineState.pedFlags (Firestore)
3	ped-02-transfer	Lesson JSON (PED-01 cleared)	PASS/FAIL verdict + mission_rewrite + verification_rewrite (if needed)	Hard FAIL → authoritative rewrite (no Content Agent needed for rewrites)	payloads/{id}/draft updated if rewrite
4	ped-04-motivation	Lesson JSON (PED-02 cleared)	PASS/SOFT_FAIL verdict + motivation_score + controlling_language_found	Soft FAIL → log flag, continue	pipelineState.pedFlags
5	brand-agent	Lesson JSON (PED-04 cleared)	Edited content + changelog + brand_compliance_score (0–100)	Brand FAIL (banned word or score <80) → Hard FAIL → route to Content/Brand Agent	payloads/{id}/draft + brandComplianceScore
6	assignment-agent	Lesson JSON (Brand cleared)	mission_validated + mission details + verification + flags	Hard FAIL → route to Content/Assignment Agent	payloads/{id}/draft if rewrite
7	media-agent	Lesson JSON (Assignment cleared)	media_required (boolean) + media_brief or null	Advisory only — never blocks	payloads/{id}/draft (media_brief)
7a	ped-07-accessibility (Premium)	Lesson JSON (Media assessed)	accessibility_score + issues + plain_language_score + alternative_paths_needed	Soft FAIL (log). Hard FAIL (paid tool, no alt path)	pipelineState.pedFlags
8	ped-06-assessment	Lesson JSON (all checks cleared)	alignment_score + issues + verification_rewrite (if needed)	Hard FAIL → authoritative Verification rewrite	payloads/{id}/draft if rewrite
9	qa-reviewer	Final lesson JSON	SOP-05 score (0–100) + verdict + per-criterion scores + target_agent + revision_prompt	Publication gate: PASS+ → sync chain. FAIL → target_agent (max 2 retries)	payloads/{id}/qaScore + qaVerdict
10	csv-qa-sync	QA Detail + LessonID	Lesson_QA_Records.csv updated + Lessons.csv Status updated	ABORT on malformed input	Sheets: Lesson_QA_Records tab
11	csv-modules-sync	QA sync confirmation + ModuleID	Modules.csv: PublishedLessons,	Log discrepancy, continue	Sheets: Modules tab

Step #	Skill	Input	Output	Gate	Storage
			AvgQAScore, QAPassRate, LastUpdated		
12	csv-courses-sync	Modules.csv rollup	Courses.csv: CompletionRate, AvgQAScore, TotalLessons, PublishedLessons, Status	ABORT on invalid CourseID	Sheets: Courses tab

10.2 Workflow B — Lesson Sending Engine

Step	Skill/Component	Description	Platform	SLA
B 1	Scheduler trigger	Daily trigger fires at 08:00 per learner timezone (Scheduler.gs)	GAS installable trigger	< 1 min to trigger
B 2	DeliveryQueue query	SheetsDataSync queries Status=queued rows where ScheduledFor ≤ now	GAS + SheetDb	< 2s per learner
B 3	Lesson fetch	LmsLessonService fetches Lessons row by LessonId (Status=Live). Falls back to next Live lesson if not found.	GAS + SheetDb	O(1) TextFinder lookup
B 4	Block Kit formatting	SlackBlockKitBuilder formats 7 ULC sections + Submit Block into Block Kit JSON. Appends CanvasUrl button if present.	GAS	< 100ms per lesson
B 5	DM delivery	SlackApiClient.openDm() + postMessage(). Queue-first: append to Sheets queue, return immediately to avoid Slack timeout.	Slack Web API	< 3s total (queue-first)
B 6	State update	LearnerProgressStateMachine: queued → delivered. LearnerProgress updated. DeliveryQueue row updated.	GAS + SheetDb	< 1s per learner
B 7	Audit write	AuditLogger.log() writes entry with correlationId. Actor=scheduler.	GAS + SheetDb AuditLog	Always synchronous

10.3 Workflow C — Submission & Completion Sync Chain

Step	Component	Action	On Failure
C 1	SlackRouter	Receives /submit event. Verifies HMAC. Routes to LmsCompletionService.	SIGNATURE_MISMATCH → reject
C 2	LmsCompletionService	Looks up Learner_Progress by LearnerEmail + LessonID. Checks Status ≠ completed.	ALREADY_COMPLETED → idempotent return
C 3	LearnerProgressStateMachine	Validates and executes transition: delivered → submitted → completed.	INVALID_TRANSITION → error logged
C 4	LearnerProgress update	Writes CompletedAt, SubmittedAt, Status=completed. Stores SubmissionText if provided.	StorageError → retry via RetryQueue

Step	Component	Action	On Failure
C 5	Next lesson queueing	Queries next lesson for learner pathway. Creates DeliveryQueue row (Status=queued).	NO_LESSON_AVAILABLE → course completion check
C 6	csv-qa-sync (Step 10)	Syncs QA data to Lesson_QA_Records.csv (if QA data available).	ABORT_LESSON_NOT_FOUND → log, continue
C 7	csv-modules-sync (Step 11)	Rolls up ModuleID stats. Updates Modules.csv row.	Log discrepancy, continue
C 8	csv-courses-sync (Step 12)	Rolls up CourseID stats. Updates Courses.csv row.	ABORT_INVALID_COURSE_ID → log
C 9	Gate week check	If lesson is M03-W04, M06-W04, or M12-W04: triggers Gate Assessment delivery.	Assessment delivery failure logged; completion still recorded
C 10	Confirmation DM	SlackApiClient posts confirmation DM with next lesson preview to learner.	SLACK_API_ERROR → log; completion still recorded
C 11	Manager notification	If LessonID is Gate Week and learner has manager_id: DM to manager with outcome.	No manager_id → skip (not an error)
C 12	AuditLog write	AuditLogger writes full operation with correlationId linking all steps.	Always synchronous

10.4 Workflow D — Onboarding (Slack Workflow Builder)

Step	Actor	Action	Platform	Output
D 1	HR Manager	Triggers "Onboard New Team Member" form in Slack Workflow Builder	Slack Workflow Builder	Form collects: email, name, brand, role, pathway (optional)
D 2	Workflow Builder	Validates form. Posts webhook to GAS doPost() endpoint with learner data.	Slack → GAS	HTTP POST to GAS Web App URL
D 3	GAS: SlackSecurity	Verifies HMAC-SHA256 + timestamp (< 300s)	GAS 04_SlackSecurity.gs	PASS → continue FAIL → reject
D 4	GAS: LmsEnrollment Service	Checks Learners table. If not enrolled: creates Learner row with all fields.	GAS + SheetDb	Learner row created
D 5	GAS: Intake Assessment (pathway ≠ standard)	Delivers 10-MCQ assessment via Block Kit DM if pathway specified.	GAS + Slack	Assessment DM sent to learner
D 6	GAS: DeliveryQueue	Queues M01-W01-L01 (or pathway start lesson) for next delivery window.	GAS + SheetDb	DeliveryQueue row created
D 7	GAS: SlackApiClient	Posts Block Kit welcome DM to learner with programme overview and /learn instruction.	GAS + Slack	Welcome DM delivered to learner
D 8	GAS: AuditLogger	Writes LEARNER_ENROLLED entry with correlationId.	GAS + SheetDb	AuditLog row appended
D 9	GAS → Workflow Builder	Returns {ok:true, code:ENROLLED}. Workflow Builder posts confirmation to HR Manager.	GAS → Slack	Confirmation message to admin

10.5 Workflow E — Offboarding (Slack Workflow Builder)

Step	Actor	Action	Output
E 1	HR Manager	Submits "Offboard Learner" form in Slack Workflow Builder (email, reason)	Form submission sent as webhook to GAS
E 2	GAS: SlackSecurity	Verifies HMAC. Checks admin guard (UserRole=admin).	PERMISSION_DENIED if not admin
E 3	GAS: LmsEnrollmentService	Looks up Learner by email. Validates active status.	NOT_FOUND error if no active learner
E 4	GAS: SheetDb	Sets Learner._rowStatus=archived. Updates Learner record.	Learner archived in Sheets
E 5	GAS: DeliveryQueue	Cancels all pending DeliveryQueue rows for learner (Status=canceled).	All pending lessons cancelled
E 6	GAS: LearnerProgress	Writes offboard event to Learner_Progress (Status=offboarded, CompletedAt=now).	Offboard event recorded
E 7	GAS: Notification (manager)	If learner has manager_id: posts notification DM to manager.	Manager informed (if applicable)
E 8	GAS: AuditLogger	Writes LEARNER_OFFBOARDED entry.	Audit trail complete
E 9	GAS → Admin	Posts confirmation DM to admin with learner name, email, timestamp.	Admin confirmation received

10.6 Workflow F — Mix (Custom Lesson Sequence)

Step	Description
F 1	Admin submits Mix workflow form in Slack Workflow Builder: selects 2–8 lesson IDs from a predefined list.
F 2	Workflow Builder POSTs selected lesson IDs + learner email to GAS webhook.
F 3	GAS validates each LessonID against Lessons table (Status=Live). Rejects any non-Live lessons.
F 4	GAS builds hybrid delivery sequence in DeliveryQueue (one row per lesson, ScheduledFor = now + N minutes per lesson).
F 5	SlackBlockKitBuilder formats a playlist message showing all selected lessons and sequence.
F 6	Playlist DM posted to learner with all lesson previews and estimated time.
F 7	Scheduler delivers each lesson at the scheduled intervals. Normal submission flow applies per lesson.

10.7 Workflow G — Reinforcement (PED-03 Spaced Repetition)

Step	Description
G 1	Admin triggers Reinforce workflow for a learner (via Workflow Builder or /reinforce command).
G 2	LmsCompletionService queries Learner_Progress for completed lessons by this learner.
G 3	ped-03-retention skill analyses spacing: identifies lessons with reinforcement opportunity (5–15 lessons since original delivery).
G 4	Generates 3–5 retrieval-spaced follow-up lessons targeting the identified concepts.
G 5	Queues reinforcement lessons in DeliveryQueue with appropriate spacing intervals.
G 6	Normal delivery flow delivers each reinforcement lesson at scheduled time.
G 7	Learner completes and submits each reinforcement lesson; progress tracked normally in Learner_Progress.

11. Deployment Architecture & Diagram

11.1 UML Deployment Diagram



11.2 Script Properties — Complete Registry

 **RULE:** All Script Properties must be set in GAS Project Settings → Script Properties BEFORE the first doPost() call. Never store secrets in .gs source code.

Property Key	Required	Format/Example	Purpose	Rotation
SHEET_DB_SPREADSHEET_ID	Yes	1BxiM... (Sheets ID)	Primary operational data store. SheetDb bootstrap key.	On Sheets document change
SLACK_BOT_TOKEN	Yes	xoxb-...	Slack API bot token for all postMessage and openDm calls.	Every 90 days (zero-downtime rotation)

Property Key	Required	Format/Example	Purpose	Rotation
SLACK_SIGNING_SECRET	Yes	32-char hex	HMAC-SHA256 request validation. Update within 60s of Slack regeneration.	Every 90 days (critical: 60s window)
FIREBASE_SERVICE_ACCOUNT	Yes	JSON key as single-line string	Firebase Admin SDK key for Firestore access.	Every 180 days
GOOGLE_AI_STUDIO_KEY	Yes	AI Studio API key	Gemini API key for Google AI Studio App access.	Every 90 days
CLAUDE_API_KEY	Yes	sk-ant-...	Anthropic Claude API key for content agent calls.	Every 90 days
ADMIN_USER_IDS	Yes	U123,U456,U789	Comma-separated Slack user IDs for admin command guard.	On team change
ON_CALL_USER_ID	Yes	U... (Slack user ID)	On-call engineer for CRITICAL alert DMs.	On rotation change
SLACK_DEFAULT_CHANNEL	Yes	C... (channel ID)	Fallback channel for admin alerts.	On channel change
BACKUP_FOLDER_ID	Yes	Google Drive folder ID	Root folder for scheduled backup jobs.	On folder change
AUTHORING_SPREADSHEET_ID	Yes	Sheets ID	Lesson authoring Sheets doc (source for SheetsDataSync).	On doc change
BULLHORN_WEBHOOK_SECRET	Conditional	HMAC secret	HRIS auto-enrol webhook validation. Required if Bullhorn integration active.	Every 90 days
TEST_SPREADSHEET_ID	Dev/Test only	Sheets ID	Test environment only. Never set in production.	N/A

11.3 Deployment Procedure — Step-by-Step

Phase 1: Pre-Deployment Gates

- Checkpoint 2 (Implementation Sign-off) obtained from Platform Architect + LMS Platform Lead
- All 9 WAA design deliverables complete and reviewed
- All testing checklist items passing (Section 12)
- Backup of existing production Sheets data taken to BACKUP_FOLDER_ID
- All 13 Script Properties documented and ready to set
- Slack App credentials refreshed; Signing Secret and Bot Token confirmed valid
- Firebase service account key generated and confirmed

Phase 2: GAS Deployment

- Open Google Apps Script editor (script.google.com) using production service account
- Create new project — name: "RWR LMS Host"
- Upload all .gs files (00_WebAppEntry.gs through 21_Util.gs) maintaining numeric prefix order
- Link SheetDb library: Resources → Libraries → add SheetDb Script ID
- Set all Script Properties from §11.2 checklist via Project Settings → Script Properties
- Run validateConfig() from GAS editor — confirm no ConfigError thrown
- Run setupTriggers(): creates daily delivery 08:00 NZST, hourly reminders, weekly report Mon 09:00, health check every 30 min
- Deploy: Deploy → New Deployment → Web App → Execute as: Me → Access: Anyone with Google account
- Copy deployment URL — this is the Slack Request URL for all 3 slash commands

Phase 3: Slack App Configuration

17. Slack API Dashboard → App → Slash Commands → Update Request URL for: /learn, /submit, /progress
18. Verify all 3 commands point to new GAS deployment URL
19. Verify Bot Token Scopes: chat:write, im:write, commands, reactions:read
20. Check Event Subscriptions: reaction_added event enabled
21. Workflow Builder: confirm onboarding, offboarding, mix, reinforce, report workflows all point to new GAS URL

Phase 4: Data Seeding & Smoke Tests

22. Run bootstrap script to seed Lessons.csv data into production Sheets (Status=Live lessons only)
23. Verify all 13 Sheets tabs exist with correct column headers in canonical order
24. Run all 8 smoke tests (SMOKE-01 through SMOKE-08) — all must PASS
25. Run runHealthCheck() — confirm Sheets, GAS quota, Slack reachability all GREEN
26. Enrol pilot cohort (5 learners via /enroll)
27. Confirm pilot learners receive Day 1 welcome DM and first lesson

12. Testing Plan

12.1 Testing Strategy — RAD Iterations

i INFO: Following RAD methodology, testing is iterative. Each phase includes a testing sprint before the next phase begins. Automated tests run in the GAS test harness (19_HostTestHarness.gs) against TEST_SPREADSHEET_ID only.

Test Level	Scope	Tool	Environment	Gate
Unit Tests	Individual GAS service functions. Slash command handlers. State machine transitions.	19_HostTestHarness.gs (runAllTests())	Test Sheets (TEST_SPREADSHEET_ID)	All ✓ before integration tests
Failure Path Tests	Non-enrolled learner, already-completed lesson, duplicate enrolment, missing config.	19_HostTestHarness.gs	Test environment	All ✓ before security tests
Security Tests	HMAC validation, timestamp replay, admin guard, unknown command, missing config.	19_HostTestHarness.gs	Test environment	All ✓ before integration tests
Integration Tests	Full lesson delivery cycle, QA pipeline + csv sync chain, admin report, overdue reminder, daily delivery job.	Manual via GAS editor	Test environment	INT-01 through INT-05 all PASS
Performance Tests	SLA validation for all commands: /learn <5s, /submit <30s, /progress <3s.	Manual timing via GAS Execution Log	Test environment	All SLAs met
UAT	End-to-end user acceptance by HR Manager, LMS Admin, Learner, System Admin.	Manual testing via Slack	Staging (prod-like)	All 6 UAT scenarios signed off
Smoke Tests	8 post-deployment smoke tests after every production deployment.	Manual Slack commands	Production	All 8 PASS — rollback if any FAIL

12.2 Unit Test Matrix

Test ID	Function	Command/Trigger	Setup	Expected Result	Pass Criteria
UT-01	testLearnHappyPath	/learn	U_TEST_LEARNER enrolled in COURSE_12M, M01-W01-L01 queued	{ok:true, code:LESSON_DELIVERED}	Block Kit DM received; Learner_Progress=delivered; AuditLog entry written
UT-02	testSubmitHappyPath	/submit M01-W01-L01 complete	U_TEST_LEARNER has M01-W01-L01 in delivered state	{ok:true, code:LESSON_COMPLETED}	Learner_Progress=complete; next lesson queued; confirmation DM sent
UT-03	testEnrollHappyPath	/enroll ops@rwrgroup.com	U_NEW_LEARNER not in Learners table	{ok:true, code:ENROLLED}	Learner row created; DeliveryQueue row queued; welcome DM sent
UT-04	testProgressHappyPath	/progress	U_TEST_LEARNER has 3 completed lessons	{ok:true}	Progress summary Block Kit posted; shows correct counts

Test ID	Function	Command/Trigger	Setup	Expected Result	Pass Criteria
UT-05	testAlreadyCompletedIdempotent	/submit (second click)	Lesson already Status=completed	{ok:true, code:ALREADY_COMPLETED}	No duplicate Learner_Progress row; original completion preserved
UT-06	testAlreadyEnrolledIdempotent	/enroll (enrolled learner)	Learner already in Learners table	{ok:true, code:ALREADY_ENROLLED}	No duplicate Learner row; original enrolment preserved
UT-07	testLearnNotEnrolled	/learn	U_NOT_ENROLLED not in Learners	{ok:false, code:ENROLLMENT_NOT_FOUND}	Error DM sent; no Learner_Progress row created
UT-08	testReportAdminOnly	/report	U_TEST_LEARNER (UserRole=learner)	{ok:false, code:PERMISSION_DENIED}	No report generated; error message sent
UT-09	testHMACInvalidSignature	Any command	Wrong X-Slack-Signature header	{ok:false, code:SIGNATURE_MISMATCH}	Request rejected; no data written; AuditLog entry written
UT-10	testTimestampReplay	Any command	Timestamp > 300 seconds old	{ok:false, code:TIMESTAMP_INVALID}	Request rejected; replay attack blocked
UT-11	testMissingConfig	Any command	SLACK_BOT_TOKEN removed from Script Properties	ConfigError thrown → CONFIG_ERROR response	No data written; error logged to Execution Log
UT-12	testAuditLogWritten	/learn	U_TEST_LEARNER enrolled	AuditLog row count increases ≥ 1	AuditLog.Actor=learner email; Action=LESSON_DELIVERED; ResourceId=LessonID

12.3 Integration Test Scenarios

ID	Scenario	Steps	Pass Criteria
INT-01	Full lesson delivery cycle	Enrol learner → runDailyLessonDelivery() → /submit → check Learner_Progress	Learner_Progress=completed; AuditLog has ≥ 3 entries with same correlationId
INT-02	QA pipeline + csv sync chain	Run Content Agent → QA Reviewer STRONG_PASS → csv-qa-sync → csv-modules-sync	Lesson Status=Live; Modules.csv PublishedLessons incremented; all 3 CSVs consistent
INT-03	Admin report generation	/report as admin → confirm Slack DM	Block Kit message delivered; report data matches Sheets totals within ±1
INT-04	Overdue reminder dispatch	Set Learner_Progress Status=overdue → run runHourlyReminderCheck()	Reminder DM dispatched to learner; AuditLog entry written; no duplicate DMs
INT-05	Daily lesson delivery job	Trigger runDailyLessonDelivery() manually → confirm lessons delivered	All enrolled learners with queued lessons receive DM; no duplicates; AuditLog complete

12.4 UAT Scenarios

Test ID	Scenario	Tester Role	Steps	Acceptance Criteria
UAT-01	New employee onboarding flow	HR Manager	Submit onboarding form → confirm learner receives welcome DM and first lesson	All 3 actions complete without error; welcome DM received within 60s; no developer involvement required

Test ID	Scenario	Tester Role	Steps	Acceptance Criteria
UAT-02	Learner progresses through 3 lessons	Learner	/learn × 3 → /submit × 3 → /progress	Each lesson delivered correctly; progress updates after each submit; /progress shows accurate count
UAT-03	Admin weekly completion report	HR Manager	/report as admin → review output	Report Block Kit delivered within 10s; completion data matches AppSheet Learner view
UAT-04	Admin enrolls learner without developer help	HR Manager	Submit onboarding form in Slack Workflow Builder	Enrolment completes; learner receives welcome DM; admin receives confirmation — zero developer actions needed
UAT-05	Overdue reminder behaviour	Learner (simulated)	Enrol learner → do not submit for 3 days → check reminder received	Reminder DM received; contains lesson title and days overdue; tone is supportive not shaming
UAT-06	AppSheet admin console	LMS Administrator	View Learner management tab → approve a lesson status → check system status panel	Lesson status updates in Sheets within 30s; system status panel shows current health; no write conflicts

12.5 Performance SLA Matrix

Command / Job	Target	WARNING Threshold	CRITICAL Threshold	Measurement Point	Action on CRITICAL
/learn (lesson delivery)	< 5s end-to-end	> 8s	> 15s	doPost() entry to Slack DM confirmed	Alert #lms-ops-alerts; check GAS quota; review Slack API latency
/submit (QA + sync chain)	< 30s end-to-end	> 45s	> 90s	doPost() entry to csv-modules-sync complete	Check RetryQueue; verify Sheets write performance; review sync chain
/progress (summary)	< 3s end-to-end	> 5s	> 10s	doPost() entry to Slack DM confirmed	Check SheetDb TextFinder performance; review Learner_Progress table size
/help	< 2s	> 3s	> 5s	doPost() entry to ContentService response	Check script execution time; review SlackService handler
runDailyLessonDelivery()	< 5 min total (full cohort)	> 8 min	> 15 min	Trigger fired to last DM confirmed	Split cohort into batches; reduce DeliveryQueue query size
runHourlyReminderCheck()	< 2 min total	> 4 min	Abort if > 8 min	Trigger fired to last DM confirmed	Check Learner_Progress table size; add overdue filter index
runHealthCheck()	< 30s	> 60s	> 120s	Trigger fired to HealthLog entry	Reduce health check scope; check Sheets API quota

12.6 Post-Deployment Smoke Test Checklist

#	Test	Action	Pass Criteria	Result
SMO KE-01	Bot is alive	/help in any Slack channel where bot is present	Bot responds with commands list within 5 seconds	☐
SMO KE-02	HMAC validation active	Send test request with bad X-Slack-Signature	{ok:false, code:SIGNATURE_MISMATCH} — no data written	☐

#	Test	Action	Pass Criteria	Result
SMO KE-03	Lesson delivery works	/learn as test learner account	Lesson Block Kit DM received in learner DM within 8 seconds	☐
SMO KE-04	Submission + sync chain	/submit {lesson_id} complete after SMOKE-03	Learner_Progress=completed in Sheets; AuditLog entry written	☐
SMO KE-05	Admin auth guard	/report as non-admin test user	{ok:false, code:PERMISSION_DENIED} — report NOT generated	☐
SMO KE-06	Admin report works	/report as admin user	Completion report Block Kit posted to admin DM within 10 seconds	☐
SMO KE-07	Audit logging active	Any command from SMOKE-03 or SMOKE-04	AuditLog tab has ≥ 1 new row with correct correlationId	☐
SMO KE-08	Scheduled trigger exists	Check GAS Triggers panel (My Triggers)	≥ 4 active triggers: daily delivery, hourly reminders, weekly report, health check	☐

⚠ WARNING: If ANY smoke test fails after deployment: execute rollback immediately. Step 1: notify System Administrator and LMS Platform Lead via #lms-critical. Step 2: GAS editor → Deploy → Manage Deployments → previous version → Set Active. Step 3: revert any changed Script Properties. Step 4: restore Sheets backup if data written. Recovery Time Target: < 15 minutes.

13. Project Management Plan

13.1 RAD Phase Breakdown — Work Breakdown Structure

Phase 1: Requirements Planning (Weeks 1–2)

Work Package	Owner	Effort (days)	Deliverable
Project charter & stakeholder register	LMS Platform Lead	1	§1 of this document
Use case specifications (UC01–UC25)	Platform Architect + Content Ops	2	§3 of this document
User stories (EP01–EP14)	Platform Architect	2	§4 of this document
41-column Lessons.csv schema lock	Platform Architect	1	§8.1 canonical schema
CMC 7-step workflow execution	Content Operations	2	Final Output Package (Step 7)
Content plan: 288-lesson topic map	Content Operations	3	Course map per month M01–M12
Skills Compendium review & gap register	System Administrator	1	Missing skill identification (csv-courses-sync, ped-07)
Data model design & ERD	Platform Architect	2	§6 database schema
Stakeholder review & sign-off	All leads	1	Phase 1 gate: CMC Step 7 approved

Phase 2: User Design — Content Engine (Weeks 3–6)

Work Package	Owner	Effort (days)	Deliverable
Master System Prompt v2.0 finalization	Content Operations	2	14-agent pipeline specification
14-agent pipeline build (Steps 0–12)	Content Operations + Platform Architect	5	Full pipeline operational in AI Studio
Publish Panel build	Platform Architect	3	ULC completeness + word limit + banned word + QA verdict + test-send gates
Firebase schema migration (role-based)	Platform Architect	4	Target Firestore schema live; data migration complete
API key security migration (callLLM)	Platform Architect	3	No client-side key storage
Notion curriculum database seeding	Content Operations	10	M01–M12 lesson pages created in Notion (288 total)
Content factory run: M01 lessons (24)	Content Operations	5	24 lessons QA-approved, Status=Ready in Lessons.csv
Brand Agent + tone presets QA	Content Operations	2	All 7 tone presets tested; banned word list validated
Phase 2 gate: first STRONG_PASS lesson + Publish Panel functional	LMS Platform Lead	1	M01-W01-L01 STRONG_PASS; Publish Panel gates all working

Phase 3: User Design — Delivery Engine (Weeks 7–10)

Work Package	Owner	Effort (days)	Deliverable
GAS codebase implementation (00–21_gs)	Platform Architect	8	All 22 GAS files passing unit tests
SheetDb table schemas & validation	Platform Architect	2	All 13 tables registered; bootstrap script working
Slash command deployment (3 commands)	System Administrator	1	/learn, /submit, /progress live in staging
Slack Workflow Builder admin workflows	System Administrator	2	Onboarding, offboarding, mix, reinforce, report flows operational
AppSheet admin console configuration	System Administrator	3	All views: Learner mgmt, QA dashboard, progress, audit viewer
Unit + security + failure path tests	Platform Architect	3	All 12 UT tests passing; all 5 security tests passing
Integration tests (INT-01 through INT-05)	Platform Architect	2	All integration scenarios PASS
Performance testing & SLA validation	Platform Architect	1	All SLAs met for /learn, /submit, /progress
UAT with HR Manager + LMS Admin + Learner	LMS Platform Lead	3	All 6 UAT scenarios signed off
Lessons.csv staging load (M01 live lessons)	System Administrator	1	M01 lessons loaded; SMOKE-01–08 pass in staging
Phase 3 gate: all smoke tests PASS + UAT sign-off	LMS Platform Lead	1	Green light for Phase 4

Phase 4: Cutover & Deployment (Weeks 11–12)

Work Package	Owner	Effort (days)	Deliverable
Production GAS deployment	System Administrator	1	Web App deployed; deployment URL captured
Script Properties set in production	System Administrator	0.5	All 13 properties set and verified
Slack App updated with production URL	System Administrator	0.5	All 3 slash commands pointing to production URL
Production smoke tests (SMOKE-01–08)	System Administrator + Platform Architect	1	All 8 PASS — no rollback triggered
Pilot cohort enrolment (5 learners)	HR Manager	0.5	All 5 learners enrolled; first lesson scheduled
Day 1 delivery monitoring	System Administrator	1	All 5 pilot learners receive first lesson; no errors
RUNBOOK.html delivery	Platform Architect	2	Interactive runbook with incident response procedures
DEPLOYMENT.md delivery	Platform Architect	1	Step-by-step deployment guide for repo root
Phase 4 gate: pilot cohort active + all monitors green	LMS Platform Lead	1	Project milestone M11 achieved

13.2 Risk Register

Risk ID	Risk Description	Probability	Impact	Mitigation	Owner
R01	GAS 6-minute execution timeout during large cohort delivery	Medium	High	Queue-first pattern; split cohort into batches; monitor execution time per trigger	Platform Architect
R02	Slack API rate limiting during peak delivery windows	Low	High	Exponential backoff in SlackApiClient; RetryQueue for failed deliveries; stagger delivery times	System Administrator
R03	Firebase schema migration causes data loss in Firestore	Medium	High	Full Firestore backup before migration; staged migration in staging environment first; rollback procedure tested	Platform Architect
R04	API key security vulnerability before callLLM Cloud Function deployed	High	Critical	Immediate priority (P0); no client demos until resolved; temporary access restriction to AI Studio App	Platform Architect
R05	Content quality: fabricated statistics reach learners	Medium	High	PED agents + QA Reviewer enforce no fabricated stats; directional language rule; M01-W01 always human-reviewed	Content Operations
R06	Lesson Status=Ready but content misaligned with SlackThreadText	Low	Medium	Systematic cross-checking during QA runs; csv-qa-sync validates field alignment before write	Content Operations
R07	Google Sheets API quota exceeded during heavy load	Low	High	Monitor via runHealthCheck(); reduce batch sizes; cache frequently-read static data	System Administrator
R08	Learner in wrong timezone receives lesson at wrong time	Medium	Medium	time_zone field in Learners table; Scheduler.gs reads per-learner timezone; default NZST with AU override	Platform Architect
R09	csv-courses-sync not registered in Skills Compendium before use	High	Medium	Generate Skill Creation prompt immediately; do not run csv-courses-sync until registered	Content Operations
R10	Pilot cohort dropout due to lesson quality issues	Low	High	All M01 lessons QA-scored ≥ CONDITIONAL_PASS before pilot; human review on M01-W01; HR Manager has /gaps access	HR Manager

13.3 Change Management Protocol

Change Type	Process	Approval Required	Timeline
Lesson content change (post-publish)	Flag in Notion → run through QA pipeline → new version replaces Lessons.csv row → Status=Live again	Content Operations sign-off + LMS Admin approval	≥ 2 business days (full pipeline)
New slash command	Full WAA 9-step design process → Checkpoint 1 + Checkpoint 2 → GAS implementation → new deployment	Platform Architect + LMS Platform Lead (Checkpoint 2)	≥ 2 weeks
Skills Compendium update (new skill)	CMC Step 5 prompt → skill registered → Compendium updated	Content Operations	≤ 1 business day

Change Type	Process	Approval Required	Timeline
Schema change (Lessons.csv column add)	MetadataVersion increment → update SheetDb schema → update GAS all references → regression tests → new deployment	Platform Architect + LMS Platform Lead	≥ 1 week
Emergency hotfix (critical incident)	Incident SOP → rollback if needed → hotfix in staging → SMOKE tests → expedited production deployment	LMS Platform Lead	< 15 minutes (rollback) / < 4 hours (hotfix)
Learner pathway change	Admin updates pathway field in AppSheet → Scheduler respects new pathway on next delivery	HR Manager or LMS Admin	Immediate (next delivery window)

13.4 Definition of Done

Sprint / Phase Done:

- All unit, failure path, and security tests PASS in test environment
- All integration tests for this phase PASS
- Performance SLAs met for all commands implemented in this phase
- AuditLog writes verified for all new operations
- AppSheet admin views reflect all new data correctly
- Skills Compendium updated for any new skills introduced
- Checkpoint 1 (design) or Checkpoint 2 (implementation) sign-off obtained as applicable
- No CRITICAL defects open; all WARNING defects triaged with owners

Production Release Done:

- All 8 smoke tests PASS in production environment
- No data written to production by test scripts
- All 13 Script Properties set and verified in production
- All 4 GAS scheduled triggers active (daily, hourly, weekly, health check)
- Pilot cohort (≥ 5 learners) enrolled and receiving lessons
- RUNBOOK.html and DEPLOYMENT.md delivered to repo root
- LMS Platform Lead sign-off obtained

Appendix A — System File Manifest

File	Class / Functions	Responsibility
00_WebAppEntry.gs	doPost(), createHostDbClient(), createHostRouter()	DI factory and entry point. Bootstraps SheetDb client and GAS web app router.
01_SlackRouter.gs	SlackRouter	HTTP boundary: parse → verify → context → dispatch.
02_SlackPayloadParser.gs	SlackPayloadParser	Normalise slash commands and webhook payloads to canonical schema.
03_SlackService.gs	SlackService	Command registry — 12 slash commands with admin guards and argument validation.
04_SlackSecurity.gs	SlackSecurity	HMAC-SHA256 verification with timestamp replay protection. P0 security requirement.
05_LmsEnrollmentService.gs	LmsEnrollmentService	Handle /enroll, onboarding webhook: create Learner row, dispatch welcome DM, schedule first lesson.
06_LmsLessonService.gs	LmsLessonService	Handle /learn: fetch lesson, format Block Kit, post DM, update progress.
07_LmsProgressService.gs	LmsProgressService	Handle /progress: query Learner_Progress, format summary, post DM.
08_LmsCompletionService.gs	LmsCompletionService	Handle /submit and  reaction: record completion, trigger csv sync chain.
09_LmsReminderService.gs	LmsReminderService	Detect overdue learners, dispatch nudge DMs on hourly schedule.
10_LmsReportService.gs	LmsReportService	Handle /report and /gaps: generate and post formatted admin reports.
11_SlackApiClient.gs	SlackApiClient	UrlFetchApp wrappers for Slack Web API: postMessage, openDm, updateMessage, lookupByEmail.
12_SlackBlockKitBuilder.gs	SlackBlockKitBuilder	Formats Block Kit JSON for all lesson delivery, progress summaries, quiz/assessment blocks, admin notifications.
13_LearnerProgressStateMachine.gs	LearnerProgressStateMachine	State transitions: queued → delivered → started → submitted → completed / overdue. Enforces valid transitions only.
14_Scheduler.gs (functions)	runDailyLessonDelivery, runHourlyReminderCheck	Daily lesson delivery; hourly overdue check; timezone-aware per-learner delivery window.
15_SheetsDataSync.gs	SheetsDataSync	Reads approved lessons from authoring Sheets; syncs Modules aggregate stats after each QA run.
16_Scheduler.gs (functions)	setupTriggers, runWeeklyReport	GAS installable triggers: daily, hourly, weekly report Mon 09:00, health check every 30 min.
17_SheetsDataSync.gs	SheetsDataSync	Bridge: reads approved lessons from authoring Sheets, upserts to delivery runtime SheetDb.
18_ConfigBootstrap.gs	ConfigError	Startup validation: checks all required Script Properties on doPost(). Throws ConfigError if missing.
19_HostTestHarness.gs (functions)	runAllTests, testLearnHappyPath, etc.	Manual test functions — all unit, security, failure path, and integration tests. Run from GAS editor only.
20_Errors.gs (constants)	Error code constants	Shared error constants and error-type registry across all service files.
21_Util.gs	Util	Shared utilities: UUID generation, ISO timestamp, row/object conversion, clone.

Appendix B — Banned Words & Brand Rules

B.1 Banned Words — Zero Tolerance

 **RULE:** Any banned word in final lesson output = automatic HARD_FAIL + Brand Compliance criterion score 0 + human review required. No exceptions. No partial matches permitted.

Banned Word / Phrase	Approved Replacement	Rationale
"just a recruiter"	(omit or reframe positively)	Diminishes professional identity — core brand violation
"human resources"	"people & culture"	RWR-specific terminology requirement
"staff"	"team" or "people"	Impersonal; contradicts people-first positioning
"manpower"	"talent" or "people"	Outdated; gender-insensitive
"headcount"	"people" or "team size"	Reduces people to numbers
"leverage"	"use" or "apply"	Corporate jargon; not plain language
"synergy"	"collaboration" or "combined effort"	Overused corporate cliché
"transformative"	"significant" or specific outcome language	Vague superlative; no concrete meaning

B.2 Brand Terminology Requirements

Use This	Not This
"empower"	"help"
"specialist brands"	"divisions"
"recruiting professionals"	"recruiters"
"people & culture"	"human resources"
"team" or "people"	"staff"

Appendix C — Related Documents

Document	Reference	Description
WAA Technical Document	RWR-TECH-WAA-001 v1.5	Authoritative technical reference for workflow automation design. 9-step design process, data model, codebase, testing, deployment.
CMG Technical Document	RWR-TECH-CMG-001 v2.1	Content Mapping & Generation platform: AI Studio app, Firebase data model, 14-agent pipeline, Slack payload system.
Master System Prompt	RWR-TECH-MSP-001 v2.0	14-agent content factory governance. Agent specifications, authority hierarchy, ULC structure, SOP-05 rubric.
Skills Compendium	March 2026 — 77 skills	Complete registry of all 77 automation building blocks across 5 categories.
Unified Platform Technical Reference	RWR-TECH-PLT-001 v2.0	Single consolidated document: live vs planned content, dependency risk register, LMS feature matrix, priority roadmap.
WAA SOP	RWR-SOP-WAA-001 v1.0	Governing SOP for all workflow automation design. Parent of WAA tech doc.
CMC SOP	RWR-LMS-SOP-001 v1.0	7-step mandatory pre-pipeline workflow. Must complete before any Content Agent run.
DEPLOYMENT.md	Repo root (to be created)	Step-by-step deployment guide for engineers setting up from scratch.
RUNBOOK.html	Repo root (to be created)	Interactive operational runbook. Incident response procedures and recovery steps.

